

1 Light-sharing control for tracker-based agrivoltaics
2 under a yield floor: what learning adds and what
3 shade-response uncertainty costs

4 Quang-Vinh Dang*
5 *British University Vietnam, Hung Yen, Vietnam*

6 **Abstract**

Single-axis trackers let agrivoltaic arrays shift their daily rotation from maximum electricity towards letting light through (light sharing). We quantify its value under a yield-retention floor of 90% with a differentiable simulator of the whole system. The crop response to shade is the least certain part: a hierarchical model fitted to 428 observations from 58 field studies predicts a new study of the four staple crop types only to about 0.22 in relative yield, no better than proportional loss, so we turned the learned curves and four field trials into ensembles of plausible responses by gradient descent through the simulator. On 92 held-out crop–site pairs, light-sharing rules allowed arrays 2.1–2.9 times as dense as static arrays at equal retention and raised the land-equivalent ratio (LER) from 1.07 to 1.18–1.24 (1.27 for a neural controller); the floor was met in 69–76% of seasons. Pooled over three splits, the neural controller beat the written phenology rule by 0.044 LER, a tuned rule by 0.018 (95% interval 0.012–0.024) and a CMA-ES-optimised schedule by 0.013; training on ensembles or a biomass signal did not help. Design mattered more: a density chosen under the mean response left about 70% of pairs at the floor when another plausible response held (oracle 85–87%), a floor raised by 0.02 was as good as an ensemble-quantile rule, and neither was safe outside the ensemble. Under illustrative costs, light sharing breaks even with the open field at 70–92 USD MWh^{−1} (static arrays at 59) and a conventional plant is more profitable above 48. The simulator’s bias is uncertain for maize and soybean (53% of the upscaled 14 PWh yr^{−1}).

7 *Keywords:* Agrivoltaics, Light sharing, Solar tracking, Food–energy nexus,
8 Crop modelling, Differentiable simulation, Hierarchical shrinkage model

9 1. Introduction

10 Two of the defining challenges of this century compete for the same re-
11 source: land. The energy transition requires terawatts of new solar pho-
12 tovoltaic (PV) capacity, and ground-mounted solar farms need roughly one
13 to two hectares per megawatt. Food production must rise while crops are
14 exposed to more frequent heat waves and droughts. Warming alone is pro-
15 jected to reduce global yields of maize, wheat, rice and soybean by several
16 per cent per degree (Zhao et al., 2017; IPCC, 2021), and a new generation
17 of climate and crop models shows these impacts emerging earlier than pre-
18 viously expected (Jägermeyr et al., 2021). Croplands are also where solar
19 power potential is greatest (Adeh et al., 2019). They are flat, sunny, acces-
20 sible and close to rural electricity demand, so converting farmland into solar
21 farms is tempting and controversial at the same time.

22 Agrivoltaic (AV) systems combine crops and PV modules on the same
23 land and aim to resolve this conflict (Dupraz et al., 2011; Dinesh and Pearce,
24 2016; Weselek et al., 2019). By sharing sunlight between modules and plants,
25 a hectare can deliver more food plus energy than the two uses side by side,
26 with land-equivalent ratios (LER) well above one in field trials and mod-
27 elling studies (Dupraz et al., 2011; Amaducci et al., 2018; Valle et al., 2017;
28 Trommsdorff et al., 2021). The partial shade also changes the crop micro-
29 climate. It lowers canopy temperature and evaporative demand, conserves
30 soil moisture and improves water-use efficiency, especially in drylands (Mar-
31 rou et al., 2013b; Hassanpour Adeh et al., 2018; Barron-Gafford et al., 2019;
32 Elamri et al., 2018; Weselek et al., 2021). Agrivoltaics is therefore a can-
33 didate adaptation measure against heat and drought stress, as reviewed by
34 Mamun et al. (2022), in addition to a mitigation measure through renew-
35 able electricity. The same shade, however, also reduces the light available for
36 photosynthesis. Regulators increasingly require AV systems to retain most
37 of the reference crop yield. The German pre-standard DIN SPEC 91434, for
38 example, requires at least 66% of the reference yield (DIN, 2021), and other
39 schemes and many farmers set stricter floors. How much electricity can be
40 produced without sacrificing food security depends on the crop, the local
41 climate, the year and the growth stage.

42 Most AV designs fix the geometry of the array, i.e. the ground coverage
43 ratio (GCR, module area per ground area) and the tilt or tracking rule, from
44 generic rules of thumb. Single-axis trackers add a degree of freedom: the
45 modules can be rotated away from the sun to let light through when the crop

needs it, as demonstrated for lettuce by Valle et al. (2017), or towards the sun to maximise electricity. Choosing the rotation every day is a sequential decision problem. The value of light depends on the crop’s phenological stage, soil water and upcoming weather, and the effect of a decision appears in the harvest months later. Recent work in this journal has used evolutionary multi-objective optimisation to lay out PV arrays on farmland (Zhang et al., 2026). Machine learning for crop management has mainly used reinforcement learning with black-box simulators (Gautron et al., 2022), which would need very many simulated seasons for global, multi-decadal assessments.

Two methodological strands are relevant. Differentiable process models, which couple physical simulators with learning, allow parameters and policies to be fitted by gradient descent through the simulator (Reichstein et al., 2019; Shen et al., 2023), and we use them for both calibration and control. Decision making under deep uncertainty asks for designs that perform acceptably across plausible models and not only under the best estimate (Marchau et al., 2019), and scenario-based designs have finite-sample guarantees under convexity assumptions that we do not meet (Calafiore and Campi, 2006; Campi and Garatti, 2008); our design rules are heuristic counterparts and we compare them with a simple safety margin.

Two questions follow. First, how much of the electricity that an agri-voltaic system can produce under a yield-retention floor comes from the array density, and how much from dynamic light-sharing control of the trackers? Second, how robust is that answer to the weakest part of any such study, the crop response to shade, and does a controller learned from a simulator add anything to a well-designed hand-written rule? Answering them requires a simulator that couples the physics of light, PV, microclimate and crop growth, a protocol in which all controllers face the same design constraints, a check of the shade response against field data, and enough sites and seasons to separate real effects from noise.

This paper makes three contributions. The first is a quantification of light-sharing control under a yield floor. Light sharing is parameterised by one daily level between full backtracking and the rotation that minimises the shadow on the crop, and we compare static arrays, a seasonal schedule, a rule that follows crop phenology, a rule that reacts to heat and drought stress, and a neural controller (SALS) trained through the simulator. The second is a deployable evaluation protocol. The array density of every site is chosen from its historical seasons only, with the same rule for every method, and performance is measured on held-out sites and years, over three training seeds

84 and three spatial splits. The third is a differentiable agrivoltaic simulator
85 that supports both, coupling hourly solar geometry, single-axis trackers with
86 backtracking, row-to-row shading and PV electricity, the direct and diffuse
87 light reaching the crop, canopy temperature and reference evapotranspiration
88 under the modules, and the SIMPLE crop model (Zhao et al., 2019) extended
89 with a flowering-stage shade penalty. All components are differentiable tensor
90 operations (Karniadakis et al., 2021). The PV part is benchmarked against
91 PVGIS. The crop response to shade is learned from the open data of a meta-
92 analysis (Laub et al., 2022), checked against field trials on rice, wheat, potato
93 and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al.,
94 2023), and propagated through the simulator as an ensemble.

95 We apply the protocol to 400 crop–site pairs that represent where maize,
96 wheat, rice, soybean and potato are grown worldwide, sampled in proportion
97 to their harvested area (Yu et al., 2020) with local crop calendars (Jägermeyr
98 et al., 2021), and driven by NASA POWER daily weather for 2001–2019
99 (Sparks, 2018) under baseline and +1.5, +2 and +3 °C climates. The main
100 findings are as follows.

- 101 1. Light sharing is the main lever on electricity at a given floor, although
102 economically it pays only at high prices (item 6). At the same mean
103 yield retention of 90%, arrays with light-sharing control carry 2.1 to
104 2.9 times the module area of static arrays and produce 1.8–2.4 times
105 as much electricity (Section 3.1).
- 106 2. The best hand-written rule follows crop phenology and needs no weather
107 forecast. Rules that react to heat or drought stress are not better and
108 depend on forecast quality (Section 3.3).
- 109 3. The crop response to shade can be learned from data. A hierarchical
110 model fitted to 428 observations from 58 field studies predicts a held-
111 out study of the four staple crop types to about 0.22 in relative yield,
112 no better than proportional loss (it is better only for crop types the
113 simulator does not represent), so between-study spread is large. The
114 simulator is close to these curves for cereals and potato but milder
115 than the meta-analysis for maize and soybean, whereas one maize trial
116 is milder still; the gains for those two crops are therefore uncertain and
117 possibly overestimated (Section 3.7).
- 118 4. A learned controller adds a small gain over the best rule: 0.044 LER
119 over the written phenology rule and 0.018 over a rule tuned on the
120 same training seasons. Training it on ensembles of shade responses, or

giving it a farm-measurable biomass signal, does not help, and with the wider ensembles it did not retain its margin over the written rule in our training budget (Section 3.8).

5. The bottleneck is the uncertainty of the crop response that enters the design. A density chosen under the mean response leaves about 70% of crop–site pairs at the floor when another plausible response holds (oracle 85–87%). A rule built on the quantiles of an ensemble of responses raises this by 9 points for 0.04 LER, but a floor raised by 0.02 under the mean response performs alike at equal compliance (differences within the uncertainty), and no rule is safe outside the ensemble, so the ensemble is useful to size a margin and not to replace it (Section 3.8).
6. Under illustrative costs, light sharing breaks even with the open field at 70 to 92 USD MWh^{−1} (static arrays at 59), learning is worth about 2 kUSD ha^{−1} yr^{−1} over a tuned rule at 90 USD MWh^{−1}, and a conventional plant replacing the crop is more profitable above its break-even of 48 USD MWh^{−1} (Section 3.9).
7. At the yield floor, agrivoltaics on 5% of the world’s staple cropland could supply about 14 PWh yr^{−1}, but only with about 10 TWp of modules, several times the installed PV capacity of the world (Section 3.11).

2. Materials and methods

2.1. Global cropland sites, weather and crop calendars

We studied five staple crops that together cover about 670 Mha of harvested area and 580 Mha of physical cropland: wheat, rice, maize, soybean and potato. Harvested areas (rainfed and irrigated) were taken from the SPAM 2010 gridded production maps (Yu et al., 2020) and aggregated to 1° cells. For each crop, 80 cells with at least 5000 ha of the crop were drawn without replacement with probability proportional to harvested area. An unweighted average over the sample therefore approximates an area-weighted average over global production. The 400 crop–site pairs lie in 381 distinct cells in 50 countries (Fig. 2). Sowing and maturity dates came from the GGCM Phase 3 crop calendar at 0.5° (Jägermeyr et al., 2021). For wheat we used the winter-wheat calendar unless the coldest-month mean temperature was below −8°C, in which case the spring-wheat calendar was used. Daily weather for 2001–2020 was retrieved from NASA POWER (Sparks, 2018): global and diffuse shortwave irradiance, maximum, minimum and

dew-point temperature, precipitation and wind speed at 2 m. Each site contributes 19 seasons (sowing years 2001–2019), each simulated as the 365 days after sowing, which gives 7600 site-seasons.

Climate change was represented by the global warming levels +1.5, +2 and +3 °C relative to 2001–2020. Daily maximum, minimum and dew-point temperatures were shifted uniformly, which keeps relative humidity approximately constant and raises the vapour-pressure deficit. The atmospheric CO₂ concentration was set to 390, 450, 500 and 600 ppm for the baseline and the three warming levels. Precipitation and irradiance were left unchanged. This delta approach isolates the thermal signal of warming; its limitations are discussed in Section 4.5.

2.2. A differentiable agrivoltaic simulator

The simulator advances all site-seasons in parallel with a daily time step for the crop and an hourly time step for radiation and PV (Fig. 1). All equations are implemented as PyTorch tensor operations (Paszke et al., 2019), so that yields and electricity are differentiable with respect to the design and control variables.

Hourly irradiance. Daily global and diffuse irradiance were disaggregated into hourly values with the Collares-Pereira and Rabl (1979) and Liu and Jordan (1960) ratios and renormalised to the daily totals. Beam irradiance on the horizontal is the difference between the two. Solar position follows the standard declination and hour-angle equations.

Array geometry and PV electricity. We consider elevated horizontal single-axis trackers with north–south axes. The axis height is 1.25 module widths, e.g. 2.75 m for the common 2.2-m single-module-high (1P) tracker. The rotation ϕ is limited to $\pm 60^\circ$. The ground coverage ratio g is the module width divided by the row pitch and is restricted to $g \leq 0.5$. For 2.2-m modules this keeps a pitch of at least 4.4 m, and a free strip of at least 2.2 m between the rows when the modules are horizontal, which is the stow position during field operations. Let ψ be the solar zenith angle projected onto the east–west vertical plane. The energy-optimal rotation with backtracking (Lorenzo et al., 2011) is $\phi_{\text{bt}} = \psi - \text{sgn}(\psi) \arccos(\cos \psi / g)$ when $\cos \psi < g$, and $\phi_{\text{bt}} = \psi$ otherwise. The *light-sharing* rotation ϕ_{sh} is the feasible angle closest to edge-on to the sun, which minimises the shadow on the crop. Given a light-sharing level $u \in [0, 1]$ chosen once per day, the rotation in every hour is

$$\phi = (1 - u) \phi_{\text{bt}} + u \phi_{\text{sh}}. \quad (1)$$

Plane-of-array irradiance combines three terms: beam irradiance on the module, reduced by the fraction of the module shaded by the neighbouring row, $\max\{0, 1 - \cos \psi / (g |\cos(\phi - \psi)|)\}$; isotropic sky diffuse; and ground-reflected light (albedo 0.2). The cell temperature follows the NOCT model (45 °C), with an hourly air temperature interpolated sinusoidally between the daily extremes. Electricity per unit land is plane-of-array irradiance $\times \eta \times g$, with $\eta = 0.21 [1 - 0.0035(T_{\text{cell}} - 25)] (1 - 0.14)$ for module efficiency, temperature losses and system losses. The installed capacity is 2.1 g MWp ha⁻¹. A conventional ground-mounted PV plant ($g = 0.4$, backtracking, no crop) is the energy reference E_{PV} .

Validation of the PV model. We compared the simulated specific yield of an isolated tracker row ($g = 0.05$, negligible row-to-row shading) with PVGIS 5.2 (European Commission, Joint Research Centre, 2022), for a horizontal north-south single-axis tracker with 14% losses, at 30 randomly chosen sites (29 returned data). Across sites, the simulated and PVGIS specific yields were correlated with $r = 0.971$. The simulator was on average 4.9% lower (mean absolute deviation 5.5%). This is consistent with its $\pm 60^\circ$ rotation limit and with the different irradiance databases (NASA POWER vs. SARA2, ERA5 and NSRDB). The benchmark covers an isolated row at low density only; electricity at high density, with row-to-row shading and diffuse-light effects, was not validated.

Light and microclimate at the crop. Averaged over the row pitch, the fraction of ground reached by beam light is $1 - \min\{1, g |\cos(\phi - \psi)| / \cos \psi\}$. The fraction of sky diffuse reaching the ground is a two-dimensional isotropic sky-view factor. We pre-computed it for 25 values of g and 13 rotation angles by ray casting over 31 neighbouring rows, and interpolate it bilinearly (differentiably). The daily ratio s of the irradiance at the crop to that of the open field defines the crop's shading. The canopy maximum temperature is lowered by $\kappa(1 - s)$.

κ is uncertain. Field studies under PV modules report small reductions of daytime crop or canopy temperature and little change in air temperature (Marrou et al., 2013b; Barron-Gafford et al., 2019), and heat-stress thresholds calibrated on air temperature do not transfer directly to canopy temperature (Siebert et al., 2014; Webber et al., 2018). We therefore use a moderate $\kappa = 1^\circ\text{C}$ as the main case and report $\kappa = 0$ (no cooling) and $\kappa = 2^\circ\text{C}$ for all methods (Section 3.2). Nocturnal warming under the modules is not

227 represented. The reference evapotranspiration under the modules, ET_0 , is re-
 228 computed with the FAO-56 Penman–Monteith equation (Allen et al., 1998)
 229 from the reduced radiation and canopy temperature. Soil water conserva-
 230 tion under the modules (Hassanpour Adeg et al., 2018; Elamri et al., 2018)
 231 therefore emerges from the model, but rain redistribution by the modules
 232 (Marrou et al., 2013a) is ignored.

233 *Crop growth.* Crop growth follows the SIMPLE model (Zhao et al., 2019)
 234 with the species parameters of its Table 1a (Table 1). Daily biomass growth
 235 is

$$\Delta B = R_{\text{crop}} \cdot \text{RUE} \cdot f_{\text{Solar}} \cdot f_{\text{CO}_2} \cdot f_{\text{Temp}} \cdot \min\{f_{\text{Heat}}, f_{\text{Water}}\}, \quad (2)$$

236 where $R_{\text{crop}} = sR$ is the radiation received by the crop. The terms are as
 237 follows.

- 238 • f_{Solar} is the canopy’s radiation interception. In SIMPLE it depends
 239 only on thermal time, with senescence accelerated by heat and drought
 240 stress. Shade therefore reduces the daily biomass increment but not
 241 leaf-area development.
- 242 • f_{Heat} decreases linearly between the heat threshold T_{heat} and the ex-
 243 treme temperature T_{ext} of the canopy maximum temperature.
- 244 • $f_{\text{Water}} = 1 - S_{\text{water}} \text{ARID}$, where the agricultural reference index for
 245 drought (Woli et al., 2012) is computed from a daily soil-water balance
 246 with curve-number runoff, drainage and root water uptake. Irrigated
 247 fields have $f_{\text{Water}} = 1$.

248 SIMPLE has a fixed harvest index and therefore ignores the high sen-
 249 sitivity of grain number to shading around flowering (Fischer, 1985; An-
 250 drade et al., 1999). It also ignores the differences in shade tolerance be-
 251 tween crop types. In the meta-analysis of Laub et al. (2022), maize and
 252 grain legumes lost yield strongly even at low shade levels, whereas C3 ce-
 253 reals and tuber crops initially lost less than proportionally; shading trials
 254 also show marked effects of shade on maize development (Ramos-Fuentes
 255 et al., 2023). We added a stage-dependent penalty. During a critical window
 256 ($0.45 \leq \text{TT}/T_{\text{sum}} \leq 0.65$), the harvest index is multiplied by $1 - h\bar{d}$, where $\bar{d}$
 257 is the mean relative radiation deficit $1 - s$ in that window. The sensitivity h
 258 is highest for maize (0.6) and soybean (0.5), the crop types that Laub et al.
 259 (2022) found most shade-sensitive, lower for the C3 cereals (wheat 0.4; rice

0.5, for which no specific curve was available) and lowest for potato (0.2). These values are assumptions that reproduce this ranking; they are not calibrated estimates. Section 3.6 shows the resulting shade-response curves. Because RUE is linear, SIMPLE does not credit the higher light-use efficiency of shaded canopies under more diffuse light. The penalty, conversely, makes shading around flowering more damaging than proportional. The net effect is closer to the field evidence than either assumption alone.

The thermal-time requirement to maturity was calibrated for every site as the mean thermal time between the GGCM sowing and maturity dates under the 2001–2020 climate, and the canopy parameters were scaled accordingly. The cultivar is thus adapted to the present climate, and warming shortens the season. Growth stops smoothly once the thermal-time requirement is met. Yield is biomass multiplied by the (shade-adjusted) harvest index. Each site is represented by a rainfed and an irrigated sub-field in proportion to its SPAM irrigated fraction. Each sub-field carries its own array, so both yield *and* electricity are area-weighted over the two sub-fields.

2.3. SALS: simulation-trained adaptive light sharing

SALS learns a *control network* $u_t = \pi_\omega(\mathbf{x}_t; g)$ (Algorithm 1). It is a multilayer perceptron with two hidden layers of 64 units and a bounded-logit output, $u = \sigma(5 \tanh(z/5)) \in [0.007, 0.993]$, which keeps gradients non-zero at the extremes. It chooses the daily light-sharing level of Eq. (1). Its inputs are the information available on day t :

- the crop and soil state: relative thermal time TT/T_{sum} , radiation interception f_{Solar} , relative soil water and the previous day’s ARID;
- a *forecast* of the day’s maximum temperature (relative to T_{heat} and T_{opt}), irradiance and ET_0 ;
- static information: the irrigation status of the sub-field, the array’s GCR g , latitude, day of season and the crop.

Forecasts are perturbed with random errors, independently every day: Gaussian with a standard deviation of 2°C for maximum temperature, and multiplicative with standard deviations of 20% for irradiance and 15% for ET_0 . The physics always uses the true weather. The same forecast errors are applied to the rule-based controllers. The controller is conditioned on g . During training, each season is simulated with a GCR close to the density

that is sustainable at its training site. This is the largest value on a coarse grid $(0.05, 0.10, \dots, 0.50)$ at which seasonal sharing keeps the mean retention of the site’s training seasons at or above ρ , plus a uniform perturbation in $[-0.05, +0.10]$ truncated to $[0.05, 0.50]$. The network therefore learns to control the densities that design selection will later pick. Outside the cropping season the trackers backtrack for maximum electricity.

For site-season i with open-field yield Y_i^{open} , let $r_i = Y_i/Y_i^{\text{open}}$ be the yield retention and $e_i = E_i/E_{\text{PV},i}$ the electricity relative to a conventional PV plant. Training minimises the penalised objective

$$\mathcal{L}(\omega) = \mathbb{E}_{i \sim \mathcal{B}} \left[-e_i + \mu_1 \max\{0, \rho - r_i\} + \mu_2 \max\{0, \rho - r_i\}^2 \right], \quad \rho = 0.9, \mu_1 = 5, \mu_2 = 30, \quad (3)$$

which combines an exact (hinge) and a quadratic penalty for the food-security floor ρ . Because r_i and e_i are outputs of the differentiable simulator, the exact gradient of Eq. (3) is obtained by back-propagation through the 365 daily steps of every season (back-propagation through time). This is the key difference from reinforcement learning with black-box simulators (Gautron et al., 2022), which estimates gradients from sampled returns.

Each minibatch contains 512 site-seasons, each simulated as a rainfed and an irrigated sub-field, drawn from one of the four climates in turn. Seasons with a failed open-field crop ($Y^{\text{open}} < 0.2 \text{ t ha}^{-1}$) are excluded from training and evaluation. The network is fine-tuned through the simulator with Adam (Kingma and Ba, 2015) (learning rate 10^{-3} , cosine decay, gradient-norm clipping at 1) for 300 steps. Growth-stage logistic functions are evaluated in a numerically stable form, so that gradients remain finite even under extreme heat.

Initialisation. Trained from random weights, the controller converged to sharing light throughout the cropping window ($u > 0.99$ on all days), i.e. to the seasonal-sharing rule, because the gradient of electricity with respect to u is small and of the same sign for most days of a season. We therefore initialise the network by behaviour cloning of the phenology rule of Section 2.4: binary cross-entropy against the rule’s output (soft targets 0.95 and 0.05) on about 1.7 million day-states visited by the rule in the training seasons, with 1500 Adam steps at a learning rate of 3×10^{-3} . Fine-tuning then moves the policy away from this initialisation wherever doing so improves the objective. Consequently, the gain of SALS over the phenology rule measures what fine-tuning through the simulator adds to that rule, not what can be learned without it.

Algorithm 1 SALS training

Require: training site-seasons in four climates; floor ρ ; penalties μ_1, μ_2

- 1: initialise π_ω by behaviour cloning of the phenology rule
- 2: **for** step = $1, \dots, N$ **do**
- 3: sample a climate and a minibatch $\mathcal{B}$ of site-seasons (rainfed and irrigated sub-fields)
- 4: set g_i to the site’s sustainable density plus a perturbation; initialise crop and soil states
- 5: **for** day $t = 1, \dots, 365$ **do**
- 6: $u_{it} \leftarrow \pi_\omega(\mathbf{x}_{it}; g_i)$ from state and noisy forecast if the crop is in the field, else 0
- 7: hourly rotation (Eq. 1), PV electricity, crop light, canopy temperature, ET_0
- 8: advance the soil-water balance and the SIMPLE crop state (Eq. 2)
- 9: $r_i \leftarrow Y_i/Y_i^{\text{open}}, e_i \leftarrow E_i/E_{\text{PV},i}$ (area-weighted); loss $\mathcal{L}$ (Eq. 3)
- 10: back-propagate through the season; Adam update of ω

329 *2.4. Deployable design selection and baselines*

330 A real array has one fixed density, which must be chosen before it is built.
331 Every method, SALS included, therefore uses the same *history-based design*
332 *rule*. For a site, the GCR is the largest value on the grid $\{0.05, 0.075, \dots, 0.50\}$
333 for which the method’s mean yield retention over the site’s 2001–2014 seasons
334 is at least $\rho = 0.9$. If no value qualifies, no array is built and the site
335 remains open field. A stricter *chance-constrained* variant requires $r \geq 0.9$ in
336 at least 80% of the historical seasons. The historical seasons of the test sites
337 are never used for training, and the weather of the evaluation years is never
338 used for design. We compare four controllers:

- 339 • *AV static*: backtracking trackers without light sharing ($u = 0$);
- 340 • *Seasonal sharing*: the light-sharing rotation ($u = 1$) throughout the
341 cropping season;
- 342 • *Stress rule*: an expert rule that shares light ($u = 1$) during the main
343 growth phase ($0.1 < TT/T_{\text{sum}} < 0.9$) except on days with a heat-stress
344 forecast ($T_{\text{max}} \geq T_{\text{heat}}$) or drought ($\text{ARID} > 0.5$), when the modules
345 track the sun and shade the crop;

346 • *SALS*: the learned controller.

347 The open field and a conventional PV plant without crops ($\text{LER} = 1$ each)
 348 are the references. Because the design rule is identical for all controllers, dif-
 349 ferences between methods reflect the quality of control and the array density
 350 that each controller can sustain under the food-security floor. We also com-
 351 pare the controllers at identical GCR over the whole grid (matched-density
 352 frontiers).

353 2.5. *A shade-response model learned from field data*

354 The simulator needs a crop response to shade that is supported by ob-
 355 servations. We learned one from the open data of the meta-analysis of Laub
 356 et al. (2022) (Zenodo 10.5281/zenodo.5716091, CC BY 4.0): 428 observa-
 357 tions of yield relative to an unshaded control at relative shade reductions of
 358 0–90% from 58 studies, with crop and crop type, experiment type (artificial
 359 shading or intercropping), climate zone, irradiation and shade type. Rela-
 360 tive yield is modelled as $y(s) = \exp(-\beta s^\gamma)$ with Student- t errors ($\nu = 4$;
 361 maximum a posteriori estimation, so this is a hierarchical shrinkage regres-
 362 sion with bootstrap intervals and not a full Bayesian model). We compared
 363 six models: proportional loss $y = 1 - s$; one curve for all crops; one curve
 364 per crop type without pooling; one curve per crop type with partial pooling
 365 (Gaussian shrinkage of $\log \beta$ and $\log \gamma$ towards a common value, scale 0.5);
 366 partial pooling with covariates on $\log \beta$ (standardised irradiation, subtrop-
 367 ical zone, solar-panel shade and intercropping); and gradient-boosted trees
 368 on the same inputs. Fits use L-BFGS. Models were compared by leave-one-
 369 study-out cross-validation, in which every study is held out once, which tests
 370 the prediction of a new study and not of a new observation in a known study.
 371 Intervals for the crop-type curves and covariate effects come from a cluster
 372 bootstrap over studies (150 resamples).

373 2.6. *Shade-response uncertainty: gradient calibration, robust controllers and* 374 *design*

375 The crop’s response to shade is the least certain part of the simulator,
 376 so we treat it as an unknown with a distribution. Two parameters per crop
 377 enter the simulator: a radiation-use compensation c (growth is multiplied by
 378 $1 + c(1 - s)$, s being the share of open-field irradiance that reaches the crop)
 379 and a multiplier h on the flowering-window harvest-index penalty. Because
 380 the simulator is differentiable, both can be fitted by gradient descent through

whole simulated seasons to published yield–shading observations. The observations are the relative yields at 20% and 40% shading of the data-driven curves of Section 2.5 for the five crops (median, with an error equal to the width of the 80% bootstrap interval divided by 2.56 and at least 0.04; rice takes the curve of C3 cereals, which contains no rice), the field trials on rice, wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023) (each season of each trial as a separate observation, error 0.06–0.08) and the unaffected harvest index of shaded maize (Ramos-Fuentes et al., 2023). We call the resulting ensemble E2. A wider ensemble, E3, uses the same procedure with a target error of 0.2 for the learned curves, which is the between-study spread of a new study (Section 3.7); the spread of the resulting members is smaller (standard deviation of relative yield across members 0.04–0.09 at 20% shading and 0.05–0.12 at 40% shading, against 0.02–0.04 and 0.03–0.06 for E2), because the fit also weights the field trials and the priors. An earlier ensemble, E1, was fitted in the same way to curves read by hand from the figures of Laub et al. (2022) and was used to train the robust and adaptive controllers below. Priors are $c \sim \mathcal{N}(0.4, 0.5^2)$ and $h \sim \mathcal{N}(1, 0.5^2)$, and the parameters are clamped to $[0, 2.5]$ (c) and $[0, 2]$ (h) after every step, which is why a response milder than the data cannot be lowered further. We fitted 12 ensemble members by randomised maximum a posteriori estimation: every member sees the observations perturbed by their errors and a prior draw perturbed by its width, and is optimised with Adam (40 steps, with the loss still falling slightly at the last step; 40 training-block seasons per crop and member; shading of the static arrays cached). The ensemble is a device for generating plausible responses and not a validated posterior: it has no coverage check, and for some parameters (the harvest-index multiplier of wheat and potato) its spread equals that of the prior. The design analyses use five fixed members (indices 0, 3, 5, 8 and 10 of 12), chosen in advance. The ensemble is used in three ways. All training and evaluation ran on a 4-core CPU; one training run takes about 20–25 minutes and the calibration of one ensemble about 7 minutes.

Robust and adaptive controllers. SALS-robust has the architecture and training of SALS (Section 2.3), but every training season is simulated with the response of a randomly drawn ensemble member that the controller does not observe, and the density jitter is widened to $[-0.05, +0.25]$ because feasible densities differ between members. SALS-adaptive additionally observes the realised biomass of the crop under the modules relative to an open reference plot, a quantity that can be measured on the farm by remote sensing or by

a reference plot. A feedback rule that shares light whenever this ratio falls below a target serves as the hand-written counterpart; its target (1.0) was chosen on training-block seasons under the ensemble.

Uncertainty-aware design. The design rule of Section 2.4 chooses the largest GCR whose historical retention meets the floor. Under shade-response uncertainty we choose the largest GCR that meets the floor in at least 75% of five ensemble members, and compare it with a design under the ensemble mean and with an oracle design under the true response. Evaluation of this rule is leave-one-out: the truth is the held-out member and the design uses the other four.

Stress tests. Controllers and rules are also evaluated under four fixed responses, the proportional response, the response calibrated to the meta-analysis, the response fitted to the field trials and a harsh response (proportional growth and a 1.5 times larger harvest-index penalty). The proportional and harsh responses lie outside the range of the ensemble.

2.7. Economic analysis

To judge whether the electricity gains matter economically, we computed an annual net value per hectare for each method relative to the open field: electricity revenue minus the annuity and operating cost of the installed capacity minus the loss of crop revenue (crop production costs are unchanged). The installed capacity is 2.1 g MWp per hectare (modules cover a share g of the land at 0.21 kWp m⁻²). Yield is dry matter, valued at assumed prices. A conventional ground-mounted plant (GCR 0.4) that replaces the crop is the comparator. All parameters (Table 15) are assumptions chosen to be representative, not data, and their ranges drive the conclusions; we therefore report break-even electricity prices and one-at-a-time sensitivities, and we do not monetise food security, which market crop prices do not capture.

2.8. Evaluation protocol, statistics and upscaling

Sites were split spatially: 10°×10° blocks were randomly assigned to training (70%) or test (30%), so that no test site has a training site in the same block. Training uses the 2001–2014 seasons of the training sites. Design selection uses the 2001–2014 seasons of the test sites. Evaluation uses the 2015–2019 seasons of the test sites. The same site-seasons are analysed in all four climates.

We report the following metrics:

- 454 • the mean electricity ($\text{MWh ha}^{-1} \text{yr}^{-1}$) and the relative electricity e ;
- 455 • the mean yield retention r and the production-weighted retention $\sum Y / \sum Y^{\text{open}}$;
- 456 • season-level compliance (share of seasons with $r \geq \rho$) and site-level
- 457 compliance (share of sites whose mean test-season retention is $\geq \rho$);
- 458 • the lower tail (10th percentile of r , share of seasons with $r < 0.8$);
- 459 • the land-equivalent ratio $\text{LER} = r + e$ (Dupraz et al., 2011).

460 Two aggregations are used and labelled in the text: tables report means over
 461 all test seasons of a sample (season-pooled), whereas statistical comparisons
 462 and intervals use the crop–site pair as the unit of analysis (mean of the pair’s
 463 seasons, then mean over pairs), so the same contrast can differ slightly be-
 464 tween the two (for example 0.034 and 0.038 for SALS minus the phenology
 465 rule on the main split). Samples are also labelled: the main analysis uses
 466 the 92 held-out pairs of one split and a GCR grid with step 0.025; analyses
 467 pooled over three splits use the 76 distinct pairs of every third held-out pair
 468 and a grid with step 0.05, and analyses of alternative shade responses on
 469 one split use every sixth (21 pairs) or every fourth pair (22 pairs). Statis-
 470 tical comparisons use site means as the unit of analysis. We report paired
 471 Wilcoxon signed-rank tests and cluster (site) bootstrap 95% confidence in-
 472 tervals of the mean difference. SALS was trained with three random seeds
 473 on the main split, and with one seed on each of two further spatial splits.

474 For the global estimate, every site is evaluated in the split in which it is
 475 held out. The mean electricity and yield per hectare of each crop are scaled
 476 to the *physical* area of that crop in SPAM 2010, which avoids double counting
 477 multiple-cropped land (Waha et al., 2020). We report the installed capacity
 478 (TWp), the capacity factor and a supply curve.

479 3. Results

480 3.1. Electricity and food on held-out sites and years

481 Table 2 compares the four controllers on the 92 held-out crop–site pairs of
 482 the main split (95 pairs are held out; pairs without a season whose open-field
 483 yield is at least 0.2 t ha^{-1} in all four climates are excluded) (seasons sown
 484 in 2015–2019), after each site’s density has been chosen from its 2001–2014
 485 seasons with the same rule for all controllers. By construction all methods

Table 1: SIMPLE crop parameters (Zhao et al., 2019, Table 1a) and the harvest-index shade sensitivity h added in this study. T_{sum} , I_{50A} and I_{50B} are rescaled per site to the local crop calendar (Section 2.2). Units: temperatures in $^{\circ}\text{C}$, thermal times in $^{\circ}\text{C d}$, RUE in gMJ^{-1} ; S_{CO_2} is the relative RUE increase per 100 ppm above 350 ppm.

Crop (cultivar)	h	T_{sum}	HI	I_{50A}	I_{50B}	T_{base}	T_{opt}	RUE	$I_{50 \text{ max } H}$	$I_{50 \text{ max } W}$	T_{heat}	T_{ext}	S_{CO_2}	S_{water}
Wheat (Batten)	0.4	2150	0.34	280	50	0	15	1.24	100	25	34	45	0.08	0.4
Rice (IR72)	0.5	2300	0.47	850	200	9	26	1.24	100	10	34	50	0.08	1.0
Maize (McCurdy 84aa)	0.6	2050	0.50	500	50	8	28	2.10	100	12	34	50	0.01	1.2
Soybean (Williams 82)	0.5	2350	0.40	600	200	6	27	0.86	120	20	36	50	0.07	0.9
Potato (Sebago)	0.2	2400	0.85	500	350	4	22	1.30	50	30	34	45	0.10	0.4

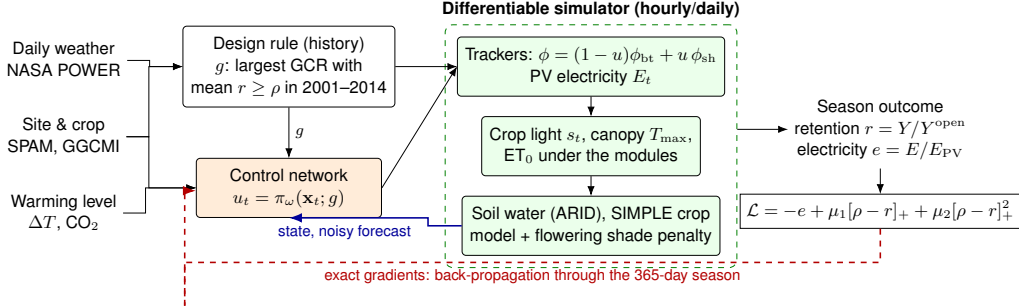


Figure 1: Simulator and controller. The array density g of a site is fixed from its historical seasons. A control network maps the daily crop and soil state, the density g and a noisy forecast to the light-sharing level u_t . The differentiable simulator converts these decisions into PV electricity, light and microclimate at the crop, soil water and crop growth. The season’s electricity and yield retention enter a penalised objective whose exact gradient is back-propagated through the 365-day season.

486 keep the mean yield retention close to 90% (0.904–0.917), so differences in
 487 electricity are differences in the density that each controller can sustain under
 488 this floor.

489 **Static arrays are the weakest.** Without light sharing, the floor limits
 490 the ground coverage to 6% on average, and arrays produce $217 \text{ MWh ha}^{-1} \text{ yr}^{-1}$,
 491 16% of a conventional PV plant per hectare (LER 1.07). Rotating the
 492 modules to let light through during the cropping season (seasonal shar-
 493 ing) allows a coverage of 17% and $397 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.19). A rule
 494 that shares light only while the crop is growing (phenology rule) reaches
 495 $465 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.24). An expert rule that additionally shades the
 496 crop on forecast heat or drought days (stress rule) is worse than the phenol-
 497 ogy rule (LER 1.18): its density is limited to 13% because it reduces light at

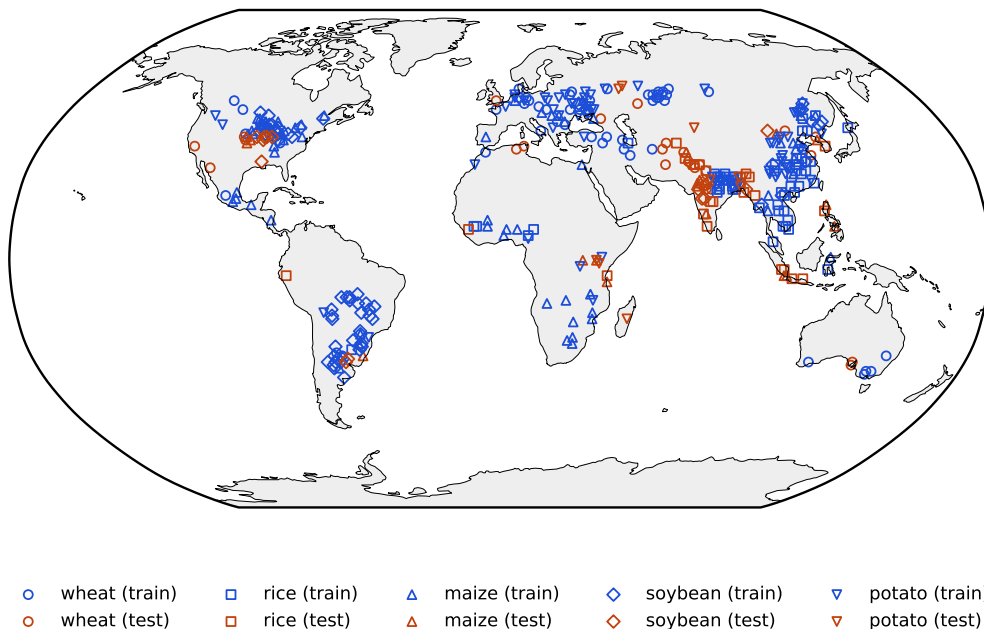


Figure 2: The 400 crop–site pairs, sampled in proportion to the harvested area of each crop (SPAM 2010). Blue: training sites; orange: geographically held-out test sites of the main split.

498 times when the floor is most binding.

499 **The learned controller is best by a modest, consistent margin.**
500 SALS sustains a density of 17% (similar to the phenology rule), produces
501 $510 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ and reaches an LER of 1.270 at the same retention. Com-
502 pared with the phenology rule this is $+0.038$ in LER (95% CI over sites
503 $0.025\text{--}0.053$; $+10\%$ electricity) and it is higher at 84% of the 92 sites (Ta-
504 ble A.18). Compared with static arrays it is $+0.205$ ($0.176\text{--}0.236$), with 2.35
505 times the electricity at 2.9 times the density. The frontier in Fig. 3 shows
506 the same picture at matched densities: the learned policy lies on or slightly
507 above the phenology rule over the whole density range, both lie well above
508 seasonal sharing and the stress rule, and static arrays are far below.

509 **What the floor does and does not guarantee.** Enforcing the floor
510 on the historical mean does not guarantee it in every season. In the test
511 seasons, 70% of SALS seasons have $r \geq 0.9$ (phenology rule 69%, stress rule
512 76%, static arrays 93%), and 2.7% have $r < 0.8$ (1.3%, 0.9% and 0.9%).

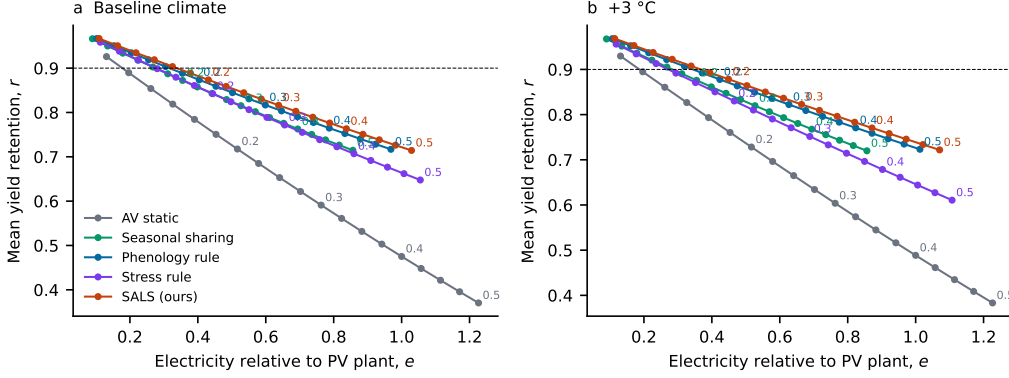


Figure 3: Electricity-retention frontiers at identical array densities (labels: GCR) for the held-out sites. Each point is the mean over test seasons at a fixed GCR; the dashed line is the 90% floor.

513 The lower tail of SALS is therefore slightly heavier than that of the rules:
514 its 10th percentile of r is 0.883, against 0.887, 0.886 and 0.903. Requiring
515 $r \geq 0.9$ in at least 80% of a site’s historical seasons (chance-constrained
516 design) raises the season-level compliance to 90–94% for the light-sharing
517 controllers. It costs about 0.065 in LER for SALS (1.205), which remains the
518 best controller (phenology rule 1.192; Section 3.2).

519 3.2. Robustness

520 **Seeds and splits.** Three training seeds of SALS give LERs of 1.270,
521 1.268 and 1.269 under the baseline climate and 1.297, 1.298 and 1.297 at
522 +3 °C. On two further spatial splits, in which different blocks of sites are
523 held out, SALS exceeds the phenology rule by 0.047 and 0.041 in LER under
524 the baseline climate and by 0.052 and 0.039 at +3 °C (main split, season-
525 pooled: 0.034 and 0.032; the site-level paired contrasts of Table A.18 are
526 0.038 and 0.034). In every split SALS ranks first, the phenology rule second
527 and static arrays last.

528 **Canopy cooling.** The canopy-cooling coefficient κ is the most un-
529 certain microclimate parameter. We repeated the whole protocol, includ-
530 ing the historical design step, with $\kappa = 0, 1$ and 2 °C. Electricity changes
531 with κ because cooler canopies tolerate denser arrays (SALS: 475, 510 and
532 549 MWh ha⁻¹ yr⁻¹ under the baseline climate), but the retention is held at
533 0.905 by the design rule and the ranking is unchanged. SALS exceeds the

phenology rule by 0.029, 0.034 and 0.042 in LER under the baseline climate (0.022, 0.032 and 0.051 at +3 °C).

Which ingredients matter. Two ablations test the design of SALS at the same training budget. Masking the stress inputs (soil water, ARID, and the maximum temperature, irradiance and ET_0 forecasts) leaves the LER unchanged (1.269 vs 1.270 under the baseline climate, 1.297 vs 1.297 at +3 °C). Training only on the baseline climate gives 1.268 and 1.291. The improvement over the rules therefore does not depend on stress-aware inputs or on training in warmed climates.

3.3. Floor conditioning, chance-constrained training and forecast errors

One network for all floors. Regulations and farmers set different floors. The network of the main experiments was trained at $\rho = 0.9$ only. We evaluated it at floors of 0.80, 0.85, 0.90 and 0.95, re-running the historical design step for each floor (Table 3). Its LER exceeds that of the phenology rule by 0.037, 0.036, 0.034 and 0.033 under the baseline climate (0.040, 0.037, 0.032 and 0.033 at +3 °C), so the benefit carries over to other floors without retraining. A network that received the floor as an additional input and was trained over all four floors did not improve on it (differences in LER ≤ 0.001). Because the design step already adapts the density to the floor, conditioning the controller on the floor is not needed.

Layout, rotation limit and harvest-index assumptions. We checked three assumptions that could drive the ranking (Table 4; 21 pairs of the main split, one training seed). The rotation limit of the trackers sets how much light sharing can deliver: with a limit of 45° the LER of the phenology rule is 1.138, with 60° 1.170 and with 80° 1.206, whereas static trackers stay at 1.05–1.06, so the benefit of light sharing grows with the limit and the ranking is unchanged. Fixed non-tracking arrays held flat or rotated by 30° about the north–south axis (facing east, not the equator; equator-facing tilts are not modelled) do no better than static trackers (LER 1.048 and 1.043 against 1.053; differences of 0.005 are within the noise of 21 pairs and one training seed), so the static tracker is not a weak baseline relative to them; vertical bifacial layouts are not modelled. The ranking of the rules also does not depend on the assumed harvest-index penalty. The window test is weakly informative about the timing of the phenology rule, which shares light through almost the whole growing period, so a shifted window cannot move it much; the informative case is the removal of the penalty. Moving the critical window to 0.30–0.50 or 0.60–0.80 of the thermal-time requirement changes the LER

571 of the phenology rule by at most 0.001, and removing the penalty altogether
572 raises all light-sharing LERs (phenology rule 1.328, SALS 1.353) without
573 changing the order: the phenology rule exceeds seasonal sharing by 0.03 to
574 0.05 in every case (0.035 with the early window, 0.034 with the late window,
575 0.053 without penalty), because it produces electricity before canopy devel-
576 opment and after maturity, when the crop does not use the light, and not
577 because of the assumed window. The margin of SALS over the written rule
578 changes with these assumptions (0.005 to 0.025 LER), and its margin over
579 the tuned rule is 0.000 to 0.009 (at a rotation limit of 45° SALS equals the
580 tuned rule).

581 **A floor of 66%.** The German pre-standard DIN SPEC 91434 requires
582 66% of the reference yield (DIN, 2021). Repeating the design step with
583 $\rho = 0.66$ (baseline climate, main split, same network trained at 0.9) the
584 ground coverage cap of 0.5 binds for the sharing methods (mean GCR 0.48
585 for the phenology rule and 0.48 for SALS, 0.44 for the stress rule), static
586 arrays reach a coverage of 0.24 and LER 1.28, the phenology rule 1.66 and
587 SALS 1.71. At this floor the density is limited by the cap and by the mean
588 retention of 0.73 reached at the cap, not by the floor, so the comparison of
589 controllers says little about control; we do not interpret it further.

590 **Chance-constrained training.** We also trained a network whose ob-
591 jective penalises a batch share of seasons below the floor that exceeds 20%.
592 It did not improve the tail. With the mean-based design step its LER was
593 1.263 (baseline) and 1.290 (+3 °C) against 1.270 and 1.297, and with chance-
594 constrained design it reached season-level compliance of 92.6% against 92.0%
595 and an LER of 1.198 against 1.205. The season-level guarantee is obtained
596 by the design step (Section 3.1), not by the training objective.

597 **Forecast errors.** We raised the forecast errors seen by the controllers
598 to 0, 1, 2 and 3 times the errors used in training. The LER of SALS and
599 of the phenology rule was identical at all four error levels to three decimals
600 (densities 0.2 and 0.3: SALS 1.307 and 1.457, phenology rule 1.278 and
601 1.416). Neither controller uses the weather forecast: both rely on the crop’s
602 thermal time, which is observable from on-farm temperature records. The
603 stress rule, which depends on the forecast of maximum temperature, degrades
604 with forecast error: at density 0.3 its retention falls from 0.778 to 0.751 and
605 its LER from 1.432 to 1.418 as the errors triple.

606 **A tuned rule.** The phenology rule above was written, not tuned,
607 whereas SALS is trained. We therefore tuned the window of the rule on
608 the training-block seasons with the same objective (electricity at the density

609 where mean retention equals the floor); the best window starts at 0.15 in-
 610 stead of 0.05 of the thermal-time requirement (Table 5). Pooled over three
 611 splits and 76 distinct crop–site pairs, the tuned rule raises LER over the orig-
 612 inal rule by 0.026 (95% cluster-bootstrap interval 0.016 to 0.037), and SALS
 613 exceeds the tuned rule by 0.018 (0.012 to 0.024), or 5.7% more electricity at
 614 equal retention, and the original rule by 0.044 (0.030 to 0.060). Tuning the
 615 rule thus accounts for about 60% of the margin of SALS over the written
 616 rule. A five-parameter ramp-plateau schedule optimised by CMA-ES on the
 617 same training seasons performs like the tuned window (LER 1.238 against
 618 1.233, difference +0.005, interval -0.001 to $+0.010$), and SALS exceeds it
 619 by 0.013 (0.007 to 0.019); the optimum is an almost binary window (start
 620 0.17, end 0.93, level 0.99), so more flexible schedules of this family do not
 621 close the gap. The remaining margin of SALS is small but not zero; it is not
 622 explained by the tuning of a window or by a smooth ramp, and we did not
 623 test richer rule families (for example schedules that depend on soil water).
 624 SALS meets the floor in fewer seasons than the rules (83% against 84–85%).

625 3.4. *What the learned policy does*

626 Averaged over the rainfed test seasons of the main split at a density of 0.3,
 627 the light-sharing level of SALS is a smooth function of relative thermal time.
 628 It is close to zero in the first 10% of the thermal time required for maturity,
 629 rises through 0.55 at 10–20%, stays at 0.94–0.99 between 30% and 70%, and
 630 falls through 0.58 at 80–90% to about zero at maturity. The plateau covers
 631 the window in which our crop model penalises shading through the harvest
 632 index. The ramps at both ends let the modules produce electricity when
 633 additional light is worth least to the crop. Fig. 4 shows two example seasons.
 634 The phenology rule, which is also the initialisation of the network, shares
 635 light during the whole growing period. Fine-tuning through the simulator
 636 found that sharing can be reduced at the edges of the season: SALS gains its
 637 advantage by yielding light when it matters and collecting electricity when
 638 it matters less.

639 The policy also shares less light on days with a forecast heat stress: the
 640 mean level is 0.35 on such days against 0.60 (baseline) and 0.51 ($+3^\circ\text{C}$)
 641 on other days, and 58–63% of sites show a lower level on heat days. This
 642 cannot be read as a stress response, however. Heat days occur late in the
 643 season, when the phenological ramp is already falling, and a policy without
 644 any stress input shows a similar contrast (0.37 vs 0.57 in a separate check).

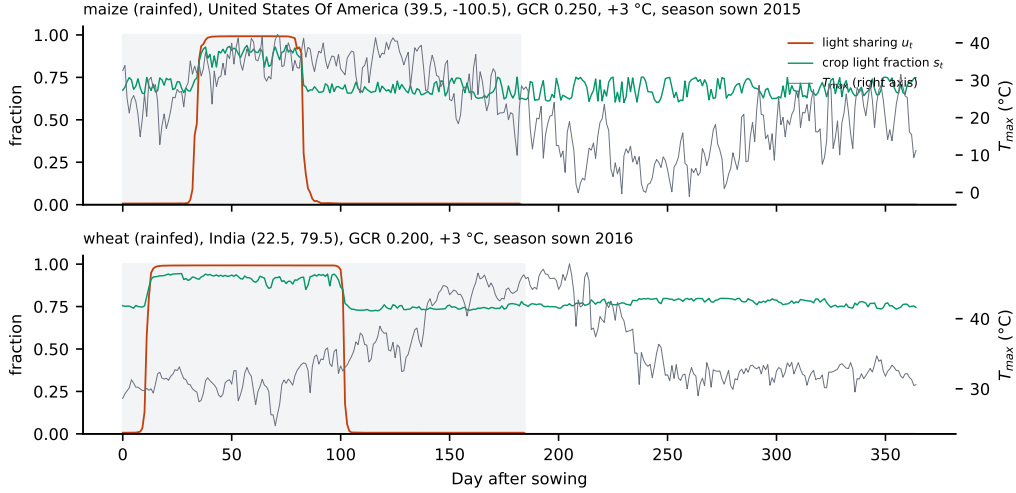


Figure 4: Learned daily control for two test seasons under $+3^{\circ}\text{C}$, at the density chosen for the site, simulated for the dominant sub-field of the site. Shading marks the cropping window. Orange: light-sharing level u_t ; green: fraction of open-field irradiance reaching the crop; grey: daily maximum temperature (right axis).

645 The ablation above confirms that stress information does not improve the
 646 result.

647 3.5. Crops and regions

648 The benefit of light sharing differs between crops (baseline climate, SALS).
 649 Soybean and maize tolerate the densest arrays (GCR 0.24 and 0.19; 691 and
 650 $577 \text{ MWh ha}^{-1} \text{ yr}^{-1}$; LER 1.41 and 1.30), potato follows (0.19; 542; 1.31),
 651 and rice and wheat are the most restrictive (0.13 and 0.15; 395 and 428; 1.20
 652 for both). The stress rule has a lower LER for every crop (1.26 for soybean,
 653 1.14–1.21 for the others).

654 The floor is not met evenly. Production-weighted retention is 0.888 for
 655 soybean and 0.898 for maize, below 0.9, because high-yielding seasons lose
 656 more than low-yielding ones. Regionally, Latin America and Oceania (4 sites)
 657 combine the highest density (GCR 0.41) and electricity ($1090 \text{ MWh ha}^{-1} \text{ yr}^{-1}$)
 658 with a production-weighted retention of only 0.862. Africa and the Middle
 659 East (8 sites) reach 0.894, and only 63% of their sites keep a mean retention
 660 of at least 0.9. South Asia, the region with the most sites (39), reaches 0.902
 661 (67% of sites) at a density of 0.14. The regional figures rest on 4 to 39 pairs
 662 without intervals and should be read as indications. A regulatory floor en-

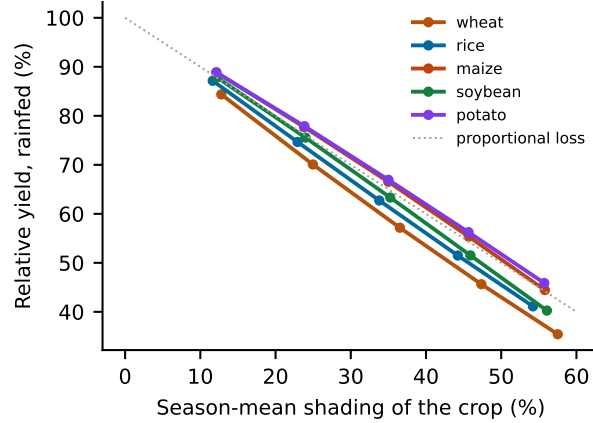


Figure 5: Simulated relative yield of rainfed crops under static arrays (GCR 0.1–0.5) against season-mean shading of the crop, baseline climate, held-out sites. The dotted line is proportional loss.

forced per site would therefore need a region- and crop-specific design step,
not a global one.

3.6. Model behaviour and validation

PV output. The simulated specific yield of an isolated tracker row agrees with PVGIS 5.2 (European Commission, Joint Research Centre, 2022) at 29 sites with $r = 0.971$, a mean bias of -4.9% and a mean absolute deviation of 5.5% (Section 2.2).

Shade response. Fig. 5 shows the simulated relative yield of rainfed crops under static arrays as a function of the season-mean shading of the crop. Yield declines close to proportionally with shading: 84–89% relative yield at 12% shading, 70–78% at 25%, 57–67% at 37% and 35–46% at 57%. Wheat loses most and potato and maize least. This response is steeper than the *less than proportional* losses that Laub et al. (2022) report for small shade levels of C3 cereals and tubers, and it does not reproduce their finding that maize and grain legumes are more shade sensitive than C3 cereals. In the simulator, shading also relieves heat and drought in rainfed maize and soybean, which offsets the higher harvest-index penalty we assigned to them. The absolute effect of shading on each crop is therefore uncertain (Section 4.5).

Comparison with field trials. We fitted one radiation-use compensation coefficient per crop to the yield–shading curves of Laub et al. (2022)

683 using training-block sites only (Methods, calibration script in the repository).
 684 The fit changed wheat and rice only (coefficients 0.4 and 0.1) because the
 685 proportional model already lies within about 0.02 of the C3-cereal and tu-
 686 ber curves, and it cannot lower the maize and legume responses below the
 687 proportional case (root-mean-square error 0.16 for maize and 0.08 for soy-
 688 bean). Against independent field trials (Table 6) the model underestimates
 689 retained yield: rice 0.79 against 0.84 (Gonocruz et al., 2021), wheat and
 690 potato 0.74 and 0.72 against 0.81–0.82 in the worse season of Weselek et al.
 691 (2021) (and above 1 in the better one), and maize 0.70–0.75 against 0.83–0.91
 692 at 30–35% shading (Ramos-Fuentes et al., 2023), where harvest index was
 693 not affected by shade. The model is therefore conservative for food reten-
 694 tion with respect to these trials, except for the stronger maize losses of the
 695 meta-analysis. Table 7 shows the consequence for the design results: with
 696 the calibrated response the allowed density and the LER of every method
 697 rise (phenology rule 1.28 against 1.23), and with a response fitted to the
 698 field trials the LER of the phenology rule is 1.62. The phenology rule is the
 699 best method in all three cases, and light sharing adds LER +0.17, +0.19 and
 700 +0.30 over static arrays (conservative, calibrated, field-fitted); only the order
 701 of the seasonal and stress rules changes. The absolute gains for rice, wheat
 702 and potato are thus conservative with respect to these trials; for maize and
 703 soybean the meta-analysis points the other way (Section 3.7).

704 3.7. A shade-response model learned from field data

705 Cross-validation by study shows what the data can and cannot support
 706 (Tables 8 and 9). Over all 339 shaded observations the root-mean-square
 707 error of a held-out study falls from 0.43 for proportional loss $y = 1 - s$ to 0.37
 708 for one curve for all crops and to 0.31 for crop-type curves, but this gain comes
 709 from crop types that the simulator does not represent (berries, forages, fruits
 710 and vegetables: 0.53 to 0.36). For the four crop types that the simulator uses,
 711 the learned curves are not better than proportional loss: the error is 0.222
 712 against 0.220 pooled (difference +0.002, 95% interval -0.023 to $+0.030$),
 713 0.187 against 0.214 for C3 cereals, and slightly worse for maize, grain legumes
 714 and tubers. Partial pooling is as good as separate curves (0.312 against 0.313)
 715 and the covariates add almost nothing (0.311). Gradient-boosted trees on the
 716 same inputs are worse (0.321) than the parametric model. The error of 0.22
 717 for the four crop types is large: yields of a new study differ from the learned
 718 curve by about 20 percentage points, which is the between-study spread
 719 that a design under uncertainty has to cover. Crop type is the information

720 that transfers between studies, and none of the covariate effects on $\log \beta$ is
721 clearly different from zero (irradiation -0.10 , 90% interval -0.32 to 0.09 ;
722 subtropical zone -0.37 , -0.73 to 0.07 ; intercropping 0.16 , -0.48 to 0.45).
723 The estimated effect of solar-panel shade is not interpretable, because all 46
724 panel observations are of leafy vegetables, mainly lettuce, and the effect is
725 confounded with crop type.

726 The learned curves show a clear ordering of crop types (Table 10). At
727 20% shading C3 cereals and tubers lose less than proportionally (relative
728 yield 0.93), whereas maize (0.73) and grain legumes (0.71) lose more; at 40%
729 shading the values are 0.72 and 0.71 against 0.42 and 0.40 . Intervals are wide
730 for maize (0.27 – 0.59 at 40%) and for tubers, which rest on two studies. The
731 simulator’s responses are close to the data for cereals and potato but, under
732 both calibrated ensembles, too mild for maize and soybean at 40% shading
733 (0.75 and 0.48 for E2 against 0.42 and 0.40 in the data). The calibration
734 cannot lower the maize response further, because the simulated shading also
735 relieves heat and drought stress and the compensation and harvest-index
736 parameters are bounded below at zero. The field trials of maize are also
737 milder than the meta-analysis: Ramos-Fuentes et al. (2023) report a harvest
738 index unaffected by shade, and Sekiyama and Nagashima (2019) a yield 5.6%
739 above the control under a low-density stilt-mounted array in one season at
740 one site. For maize and soybean the direction of the bias is uncertain, and
741 the gains reported in this paper are possibly overestimated.

742 3.8. Learning under shade-response uncertainty

743 **Calibration.** Gradient descent through the simulator fitted the five crops
744 to the data-driven curves and the field trials (ensemble E2, Table 11). The
745 data identify two features: the harvest index of maize is hardly affected by
746 shade (multiplier 0.16 ± 0.13 , $c \approx 0.14$), as observed by Ramos-Fuentes et al.
747 (2023), and soybean is more shade sensitive than the cereals (compensation
748 0.05 , harvest-index multiplier 1.66), as in Laub et al. (2022). Wheat has the
749 largest compensation (0.91 ± 0.31) and the largest uncertainty; the compen-
750 sation is an effective parameter that absorbs diffuse-light and light-saturation
751 effects that the simulator does not model explicitly, not a physical value.

752 **Learned controllers.** Controllers were evaluated under four fixed re-
753 sponses on 21 crop–site pairs of split 0 with one training seed, so the intervals
754 below, which are cluster bootstraps over pairs of per-pair mean LER differ-
755 ences, do not include seed or split variation (Table 12 gives season means,

756 which differ slightly). Under every response SALS, trained on the propor-
 757 tional response, is better than the written phenology rule: by +0.019 (0.007
 758 to 0.039), +0.024 (0.009 to 0.045), +0.038 (0.017 to 0.063) and +0.015 (0.000
 759 to 0.036) LER under the proportional, calibrated, field-fitted and harsh re-
 760 sponses. Training on the narrow ensemble E1 did not change this: SALS-
 761 robust differs from SALS by -0.005 to $+0.003$ (intervals contain zero except
 762 for the proportional response, 0.000 to 0.003), and SALS-adaptive, which
 763 sees the measured biomass ratio, by -0.010 to $+0.007$ (all intervals contain
 764 zero). Training on the wider ensembles did not help either: SALS-robust on
 765 E2 and E3 and SALS-adaptive on E3 are below SALS by 0.014, 0.017 and
 766 0.017 LER under the proportional response, 0.017, 0.021 and 0.020 under the
 767 calibrated response, 0.004, 0.005 and 0.011 under the field-fitted response (in-
 768 tervals contain zero) and 0.008, 0.014 and 0.017 under the harsh response;
 769 the intervals exclude zero in 6 of the 9 comparisons for the proportional, cal-
 770 ibrated and harsh responses. The E3 robust controller is not distinguishable
 771 from the written rule ($+0.001$ to $+0.003$, intervals contain zero) except un-
 772 der the field-fitted response ($+0.032$, 0.001 to 0.067). A tuned window and a
 773 CMA-ES ramp schedule, also evaluated under the four responses (Table 12),
 774 are 0.005 to 0.018 LER below SALS (intervals of the SALS contrast: $+0.004$
 775 to $+0.016$ against the tuned window under the proportional response, and
 776 containing zero for the ramp under the calibrated and field-fitted responses
 777 and for the window under the harsh response), and the ensemble-trained
 778 controllers are at or below the tuned window under all responses except the
 779 field-fitted one. With one seed and 21 pairs these controllers therefore did not
 780 retain the margin of SALS within our training budget; a controller that must
 781 perform across plausible responses is less specialised, but a larger budget or
 782 more seeds could change this. The feedback rule is worse than the written
 783 rule by 0.026–0.058 LER under every response (intervals exclude zero). A
 784 plausible reason is that the biomass ratio does not reveal the harvest-index
 785 loss from shading around flowering, but we did not test this explanation.

786 **Design under uncertainty.** The larger risk is in the design step (Ta-
 787 bles 13 and 14, Fig. 6). The designs below are evaluated by leaving one
 788 ensemble member out as the truth and using the other four, so that the
 789 truth never enters the design. Pooled over the three spatial splits (76 dis-
 790 tinct crop–site pairs), a density chosen under the mean response of E2 leaves
 791 70% of the pairs at the floor when the true response is another plausible one,
 792 against 87% for an oracle design (E3: 71% and 85%). Choosing the density
 793 so that the floor is met in at least 75% of the other members raises compli-

794 ance by +7.9 points under E2 (95% cluster-bootstrap interval 4.5 to 11.6)
 795 and +7.9 under E3 (3.9 to 12.4), at a cost of 0.031 and 0.025 in LER. Raising
 796 the floor used in the design from 0.90 to 0.92 under the mean response does
 797 more: +18.0 points (13.4 to 22.6) at 0.068 LER under E2 and +14.2 points
 798 (9.7 to 19.2) at 0.056 under E3, and the pairs below the floor fall short by less
 799 (mean shortfall 0.0022 against 0.0045 for the quantile rule under E2). The
 800 two kinds of rule lie on one frontier (Fig. `effig:designfrontier`). Interpolating
 801 the margin frontier (floors 0.90 to 0.96) at the compliance of the ensemble
 802 rules, the LER of the margin minus that of the rule is +0.001 (95% inter-
 803 val -0.009 to 0.012) for the rule with 75% of the other members and +0.003
 804 (-0.016 to 0.021) for the rule with all of them under E2, and -0.006 (-0.019
 805 to 0.005) and +0.024 (-0.011 to 0.058) under E3, so a simple margin and
 806 the ensemble rules are not distinguishable on this sample. The ensemble is
 807 thus useful for sizing the margin (0.02 to 0.04 above the floor for 85–97%
 808 of the pairs), not for replacing it. None of these rules is safe outside the
 809 ensemble (Table 14): under the proportional, calibrated and harsh responses
 810 the mean-response design leaves 7–17% of the pairs at the floor, a floor of
 811 0.92 or the ensemble rules 12–33%, a floor of 0.94 29–47% and a floor of 0.96
 812 51–78%, at a cost of about 0.2 LER, whereas the field-fitted response, which
 813 lies inside the range spanned by the ensembles, is handled well by every rule
 814 (80–99%) except the mean-response design (68–79%). The guarantee there-
 815 fore holds only within the range of responses that the ensemble spans, and a
 816 response harsher than the data supported by the ensemble breaks it.

817 *3.9. Economics of light sharing*

818 All figures in this section follow from the assumed parameters of Table 15
 819 and should be read as a sensitivity analysis, not a forecast. The economics
 820 are dominated by the capacity installed, not by the crop (Table 16): the
 821 annuity of the modules is 11–32 kUSD per hectare and year, whereas the
 822 crop revenue lost is about 0.3 kUSD, so a method is worth what it earns per
 823 installed watt. With the densities designed for the floor, static arrays earn
 824 +0.3 kUSD ha⁻¹ yr⁻¹ at 60 USD MWh⁻¹ and break even at 59, whereas all
 825 light-sharing methods lose money at that price (-3 to -10 kUSD) and break
 826 even at 70 (stress rule), 73 (SALS), 77 (tuned phenology rule), 80 (written
 827 rule) and 92 (seasonal sharing) USD MWh⁻¹. At 90 USD MWh⁻¹ static
 828 arrays (6.3 kUSD) are better than the written (4.0) and tuned (5.5) rules
 829 and worse than SALS (7.6), which overtakes static arrays above about 85
 830 USD MWh⁻¹. A conventional plant that replaces the crop breaks even at 48

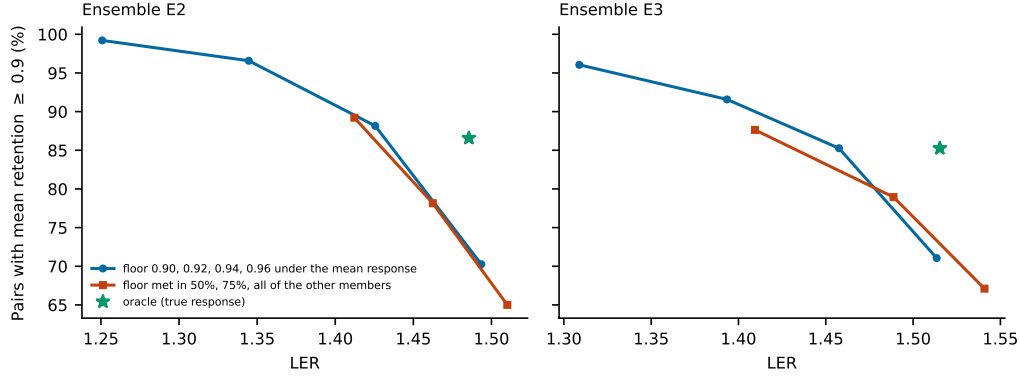


Figure 6: Share of crop-site pairs at the floor against LER for the phenology rule under design rules for shade-response uncertainty (leave-one-member-out, three splits, 76 pairs): a higher floor under the mean response (blue; the first point is the design under the mean response) and a floor met in a share of the other members (orange), with the oracle design (green).

USD MWh⁻¹ and is more profitable than every agrivoltaic option above that price (16 kUSD at 60, 94 kUSD at 120, against -6 and 21 for SALS); below it all options lose money and static arrays lose least (at 30 USD MWh⁻¹: plant -23.2, SALS -18.7, static -5.7 kUSD). Per megawatt-hour, the electricity of SALS costs about 73 USD including the crop lost, against 48 USD for the plant, a difference of 25 USD MWh⁻¹. A premium of that size would equalise the unit cost but not the profit per hectare, because a hectare of plant produces three times as much electricity (1306 against 438 MWh): relative to the plant, SALS would need a premium of about 49 USD MWh⁻¹ at a market price of 60 and about 109 at 90 to match its profit per hectare, and only about 13 at 60 (none above 73) to match the open field. The sensitivity is large (Table 17): at 90 USD MWh⁻¹ the net value of SALS changes from +17 to -8 kUSD between module costs of 0.7 and 1.5 USD Wp⁻¹, whereas crop prices matter little.

Because the density was designed for the floor and not for profit, we also let each site choose, among the densities that meet the floor in its history, the one that maximises its historical net value (sites where none pays stay open field; Fig. 7). Then no method pays at 30-40 USD MWh⁻¹; at 60 14-59% of the pairs install an array (none for seasonal sharing) and the net values are small (static arrays 0.9, SALS 0.3 kUSD); static arrays are the best option up to about 75 USD MWh⁻¹, SALS leads from about 80 (4.9

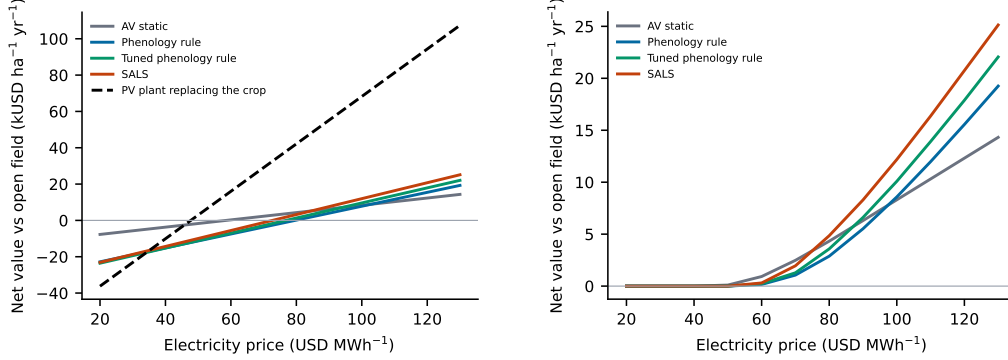


Figure 7: Net value per hectare relative to the open field against electricity price. Left: densities designed for the floor, with a conventional plant replacing the crop (dashed). Right: each site chooses the feasible density that maximises its historical net value. Illustrative parameters of Table 15.

852 against 4.3 at 80, 8.3 against 6.3 at 90, 20.7 against 12.3 at 120), and the
853 tuned rule overtakes static arrays only above about 105. The value of control
854 is therefore real but depends on a high price: at 90 USD MWh⁻¹ SALS is
855 worth 1.7 kUSD ha⁻¹ yr⁻¹ over the tuned rule with profit-optimal densities.

856 **Cost of the safety margin.** At 90 USD MWh⁻¹, moving from a density
857 designed under the mean response to a floor of 0.92 under that response costs
858 1.00 kUSD ha⁻¹ yr⁻¹ (17%) under E2 for 18 more points of compliance (70%
859 to 88%) and 1.13 kUSD (20%) under E3 for 14 points (71% to 85%), about
860 56 and 79 USD ha⁻¹ yr⁻¹ per point. The ensemble rule that requires all other
861 members to meet the floor costs 0.85 and 1.22 kUSD for 19 and 17 points
862 (45 and 74 USD per point), so on this measure it is cheaper than the margin
863 under E2 and similar under E3. At 60 USD MWh⁻¹ a margin has a negative
864 cost, because every module beyond break-even loses money; the economics
865 of the design margin therefore depend on the price level, whereas its effect
866 on compliance does not.

867 3.10. Agrivoltaics under warming

868 Under warming, the open-field production of the held-out sites (Fig. 8)
869 falls at +3 °C by 23% for maize and 18% for rice, is nearly unchanged for soy-
870 bean, and rises for wheat (+3%) and potato (+10%). Agrivoltaic production
871 with light-sharing control follows the open field at about 90% of its level, so
872 it inherits the same climate impacts (maize -31%, rice -27% relative to the

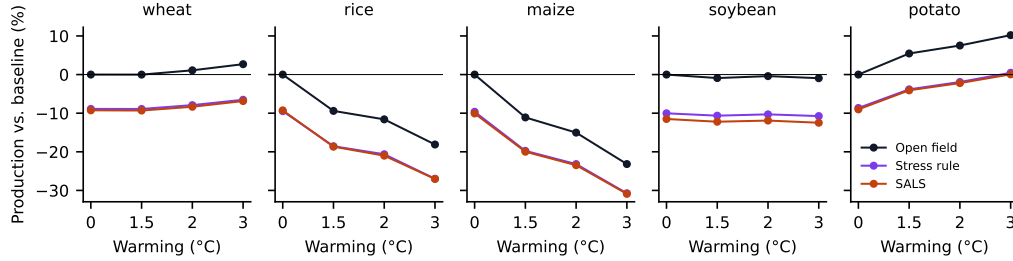


Figure 8: Production relative to the baseline open field (sum over held-out site-seasons) as a function of warming, for the open field, the stress rule and SALS.

baseline open field at +3 °C). Agrivoltaics does not protect staple production against warming in this model. The LER of light-sharing systems increases slightly with warming (SALS 1.270 to 1.297; phenology rule 1.236 to 1.265), because the floor allows slightly denser arrays in warmer conditions (GCR 0.173 to 0.177 and electricity 510 to 540 MWh ha⁻¹ yr⁻¹). The margin of SALS over the rules does not grow with warming (Table A.18).

3.11. Global potential and what it would take

Every site is evaluated when it is held out (219 of the 400 crop–site pairs, in at least one split; 222 are held out in at least one split and three have no valid season), and each crop’s mean electricity per hectare is scaled to its physical area in SPAM 2010 (581 Mha for the five crops, which avoids counting multiple-cropped land twice). Equipping 5% of this area (29.1 Mha) with SALS arrays at the floor would produce 14.4 PWh yr⁻¹ under the baseline climate and 16.0 PWh yr⁻¹ at +3 °C, with a production-weighted retention of 0.903 and 0.900. The phenology rule would give 13.0 and 14.5 PWh yr⁻¹, and static arrays 6.4 and 6.9 PWh yr⁻¹. For comparison, world electricity generation was about 30 PWh in 2023 (Ember, 2024).

These numbers require large amounts of hardware. SALS at 5% coverage needs 10.6 TWp of modules (11.3 TWp at +3 °C), at a capacity factor of 15.5% (16.2%). World solar PV capacity at the end of 2024 was about 1.9 TW (IRENA, 2025), so the requirement is about 5.7 times the installed PV fleet. The electricity is also variable, peaks at midday and would be curtailed or stored. The figures are technical potentials, not projections. Ordering sites by electricity per hectare gives a supply curve (Fig. 9): the best 2% of the staple cropland would supply 16 PWh yr⁻¹ and the best 5% 40 PWh yr⁻¹ with SALS. The best tenth of the sites are mostly rainfed maize and soybean sites

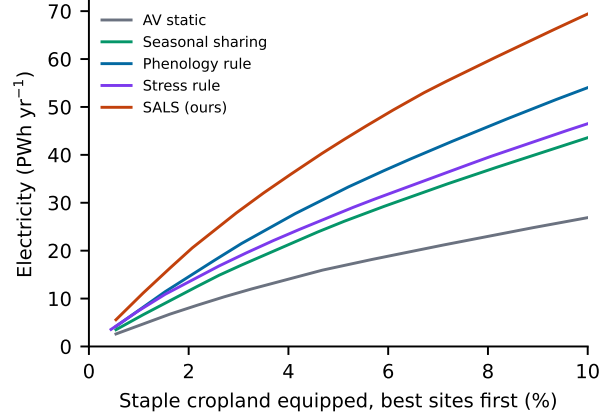


Figure 9: Supply curve under the baseline climate: cumulative electricity when sites are equipped in order of decreasing electricity per hectare, against the share of staple cropland (physical area) equipped. Each controller is evaluated at the density that keeps its mean retention at 0.9.

899 in the Americas and in southern and eastern Africa (7% mean irrigated frac-
 900 tion against 29% overall), which are also among the regions where the floor
 901 is hardest to meet (Section 3.5). Siting by electricity alone would therefore
 902 concentrate agrivoltaics where food-security risk is highest.

903 Maize and soybean, for which the simulated shade response is probably
 904 too mild (Section 3.7), account for 53% of the electricity of the upscaling
 905 (maize 30%, soybean 23%, wheat 29%, rice 15%, potato 3%). Counting only
 906 wheat, rice and potato would give about 7 PWh yr⁻¹ at the same coverage,
 907 so the global figure should be read as an upper estimate for about half of its
 908 content. By crop, the LER of SALS under the baseline climate is 1.20 for
 909 wheat and rice, 1.31 for potato, 1.30 for maize and 1.41 for soybean.

910 4. Discussion

911 4.1. What the results establish

912 Four findings stand on consistent evidence. First, sharing light with the
 913 crop is the main lever on the electricity that tracker-based agrivoltaic arrays
 914 can produce under a yield floor. With static trackers the floor limits arrays
 915 to about 6% ground coverage and an LER of 1.07. Any light-sharing con-
 916 trol allows 2.1 to 2.9 times the density, and the LER rises to 1.18–1.24 for
 917 hand-written rules. Second, the best simple rule follows crop phenology. It

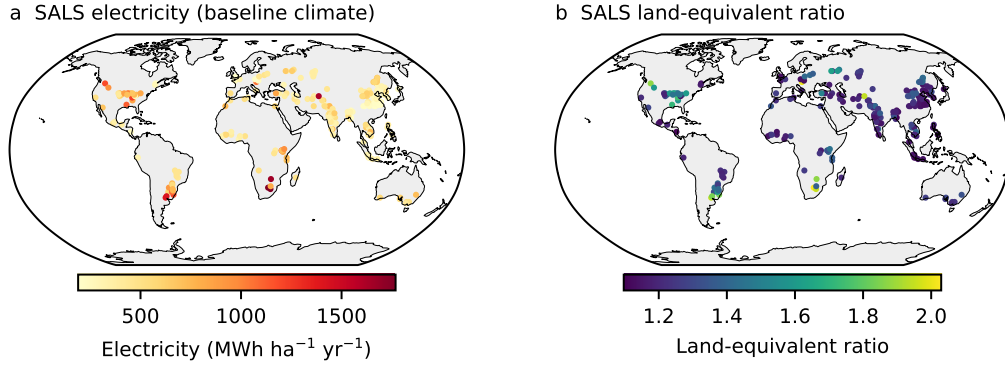


Figure 10: SALS electricity (a) and land-equivalent ratio (b) at the pooled held-out sites (baseline climate, mean of seasons 2015–2019).

918 shares light from the onset of canopy development (5% of the thermal-time
 919 requirement) until maturity and produces electricity before and after, when
 920 the crop does not use the light. It needs only the crop's thermal time, which
 921 is observable on the farm, and it is insensitive to forecast errors, whereas
 922 the rule that reacts to heat and drought stress is worse in every climate and
 923 degrades with forecast error. Third, the qualitative result does not depend
 924 on the weakest part of the model. The simulated shade response is conser-
 925 vative against the field trials we could use (Section 3.6); calibrating it raises
 926 the allowed density, and the phenology rule stays the best method, with a
 927 gain over static arrays of 0.17, 0.19 and 0.30 LER under the conservative,
 928 calibrated and field-fitted responses. However, the response learned from
 929 the meta-analysis shows steeper losses for maize and grain legumes than the
 930 simulator reproduces (Section 3.7), so the absolute gains are conservative
 931 for wheat, rice and potato and probably overestimates for maize and soy-
 932 bean (53% of the upscaled electricity). Fourth, a controller learned through
 933 the differentiable simulator is better than the best rule we could write, by
 934 0.04 in LER against the written rule (0.018 against a rule whose window
 935 was tuned on the training seasons, 5.7% more electricity at equal retention),
 936 reproducibly over training seeds (range 0.002 in LER), three spatial splits,
 937 three canopy-cooling assumptions and four climates, and at 84–86% of the
 938 held-out sites. This gain is small compared with the gap between static and
 939 dynamic control.

940 *4.2. What role does artificial intelligence play?*

941 We tested three ways in which learning could matter, and the answers dif-
942 fer. The first is learning the crop’s response to shade from data. The learned
943 model is useful mainly as a measurement of how little can be learned: a hier-
944 archical model fitted to 428 observations from 58 studies predicts a held-out
945 study of the four staple crop types with an error of 0.22 in relative yield, no
946 better than proportional loss, and the gain over proportional loss (0.43 to
947 0.31 over all types) comes from crop types that the simulator does not repre-
948 sent. Crop type is the information that transfers, whereas irradiation, climate
949 zone and shade type add nothing measurable and a non-parametric model
950 (gradient-boosted trees) is worse with this number of studies (Section 3.7).
951 Two things follow. The shade response of the staple crops is a distribution
952 with a spread of about 20 percentage points between studies and not a curve,
953 and the process-based simulator does not reproduce the steep losses of maize
954 and legumes seen in the data. Learning the response was limited by the data
955 and not by the model: two studies support the tuber curve, no rice study
956 exists, and 46 panel observations are all leafy vegetables.

957 The second is calibration through the differentiable simulator. Gradient
958 descent through whole seasons fitted the response of each crop to the learned
959 curves and to the field trials, and it produced an ensemble of plausible re-
960 sponses that reconciles them (Section 3.8). The ensemble informs the design
961 decision: a density chosen under the mean response leaves about 70% of
962 crop–site pairs at the floor if another plausible response holds, against 85–
963 87% for an oracle, and a density chosen to meet the floor in 75% of plausible
964 responses recovers about half of that gap at a cost of 0.04 LER. A simple mar-
965 gin, however, does as well: raising the floor from 0.90 to 0.92 under the mean
966 response reaches 85–88% of the pairs, and interpolating the margin frontier
967 at the compliance of the ensemble rules gives LER differences (margin minus
968 rule) of +0.001 (−0.009 to 0.012) and +0.003 (−0.016 to 0.021) under E2
969 and −0.006 (−0.019 to 0.005) and +0.024 (−0.011 to 0.058) under E3, so the
970 two are not distinguishable on this sample, and the cost per point of com-
971 pliance is similar. None of the rules is safe outside the ensemble (7–33% of
972 the pairs at the floor under the proportional, calibrated and harsh responses
973 for the mean design, the margin of 0.92 and the ensemble rules), and only a
974 floor of 0.96 reaches 51–78% at a cost of about 0.2 LER. What the ensem-
975 ble adds is the size of the margin, not a better rule. The quantile rule is a
976 scenario-type design (Calafiore and Campi, 2006; Campi and Garatti, 2008);
977 with five members it has none of the finite-sample guarantees of that theory,

978 and the comparison with the margin shows that it is not needed in practice.
 979 The bottleneck of tracker-based agrivoltaics under a yield-retention floor is
 980 the uncertainty of the crop response that enters the design, and it is data
 981 on that response, not larger networks, that would improve the result; the
 982 learned curves show a spread of about 20 percentage points between studies.
 983 The guarantee holds only within the range of responses that the ensemble
 984 spans.

985 The third is control, and here learning contributes little. The learned
 986 controller adds 0.018 LER over a tuned phenology rule (and 0.044 over the
 987 written one), and 0.015–0.038 over the written rule under the four alternative
 988 responses on one split (21 pairs); a reader who needs a simple, auditable
 989 controller can use that rule and obtain most of the electricity gain over static
 990 arrays. Neither of the two extensions that we expected to matter under
 991 uncertainty did: training on ensembles of responses (E1, E2 and E3) did not
 992 help and, for the wider ensembles, did not retain the margin over the written
 993 rule within our training budget (one seed, 21 pairs), and neither did giving
 994 the controller a farm-measurable signal, the biomass of the crop relative to
 995 a reference plot, nor a feedback rule built on that signal. The learned policy
 996 also has to be initialised from the phenology rule (Section 4.5). We therefore
 997 do not claim that learned control is what makes agrivoltaics work: in this
 998 model, the schedule that a learned policy converges to is a smooth function
 999 of crop phenology, which a rule can already express.

1000 4.3. *What the results do not establish*

1001 The early hypothesis that stress-aware control, that is, shading the crop
 1002 on heat and drought days, would be a main source of benefit under warming
 1003 was not supported. The stress rule performs worse than the phenology rule
 1004 in every climate, and a learned controller without stress inputs performs as
 1005 well as one with them. A plausible reason is that in our crop model the yield
 1006 gain from shading on a hot or dry day is smaller than the loss of light on
 1007 that day, so stress information does not help. Whether this holds for crop
 1008 models with more detailed canopy-temperature and heat-stress responses is
 1009 an open question. Agrivoltaics does not protect staple production against
 1010 warming in our simulations: light-sharing systems produce about 90% of the
 1011 open-field yield in every climate and follow the same decline in maize and
 1012 rice (Fig. 8).

1013 4.4. Food security in practice

1014 Holding the historical mean retention at 90% does not hold it in each
 1015 season or at each site. 70% of SALS seasons meet the floor, the lower tail is
 1016 slightly heavier than for the rules, and production-weighted retention is below
 1017 0.9 for soybean (0.888) and maize (0.898) and in several regions. Chance-
 1018 constrained design raises season-level compliance to about 92% at a cost of
 1019 0.065 in LER, and the ranking of the controllers is preserved. Whichever for-
 1020 mulation is chosen, a per-site regulatory floor would need a region- and crop-
 1021 specific design step. The upscaled potential (Section 3.11) shows the trade-off
 1022 in magnitude. The 14 PWh yr⁻¹ that 5% of the staple cropland could supply
 1023 need about 10 TWp of modules, several times the present global fleet, and
 1024 the best sites are rainfed maize and soybean areas where production-weighted
 1025 retention is lowest.

1026 4.5. Limitations

- 1027 • *Simulation only.* All results come from a simulator. PV output was
 1028 benchmarked against PVGIS (29 sites, mean bias −4.9%), but the crop
 1029 response to shade was learned from one meta-analysis data set and
 1030 compared with four field trials (Sections 3.6 and 3.7). The simulated
 1031 response is conservative with respect to the trials (rice, wheat, potato,
 1032 maize) but too mild for maize and soybean relative to the meta-analysis,
 1033 where the 40%-shading relative yields are 0.42 and 0.40 against 0.75
 1034 and 0.48 in the simulator. The harvest-index sensitivities h are as-
 1035 sumptions, and the sensitivity analyses use subsamples of crop-site
 1036 pairs (22 for the rule comparison under alternative responses, 21 for
 1037 the controller comparison, one training seed).
- 1038 • *Canopy cooling.* The effect of shade on canopy temperature was repre-
 1039 sented by one coefficient κ , applied to a heat-stress function that was
 1040 calibrated on air temperature (Siebert et al., 2014; Webber et al., 2018).
 1041 The ranking of the controllers is robust to κ between 0 and 2 °C, but
 1042 the absolute electricity changes by about $\pm 8\%$. Nocturnal warming
 1043 under modules and rain redistribution are not modelled.
- 1044 • *Crop model.* SIMPLE has a fixed harvest index and linear radiation-
 1045 use efficiency, and its CO₂ response is that of controlled experiments
 1046 that tend to overstate field responses (Long et al., 2006). More detailed
 1047 models with grain-number formation and canopy energy balance may
 1048 favour different schedules.

- 1049 • *Layouts and models.* The baseline of static arrays is a backtracking
1050 tracker held at the energy-optimal angle; flat and 30°-tilted fixed ar-
1051 rays did no better in a check on 21 pairs, but vertical bifacial layouts
1052 (Willockx et al., 2023; Campana et al., 2021) are not modelled, and the
1053 ground coverage ratio is a coarse proxy of crop productivity (Dupraz,
1054 2023). The PV model is isotropic in the diffuse component and has no
1055 bifacial gain; it was benchmarked against PVGIS at low density only
1056 and has not been compared with a reference implementation such as
1057 pvlib (F. Holmgren et al., 2018). The rotation limit was varied between
1058 45 and 80 degrees and changes the size of the benefit but not the rank-
1059 ing; the harvest-index window is assumed to be common to all crops,
1060 and shifting it or removing the penalty did not change the ranking of
1061 the rules.
- 1062 • *Climate scenarios.* Warming was applied as uniform temperature shifts
1063 with fixed precipitation, radiation and relative humidity, and with un-
1064 changed cultivars and sowing dates. It is not a bias-adjusted climate
1065 projection.
- 1066 • *Initialisation and state information.* The network is initialised from the
1067 phenology rule, because training from random weights collapsed to full
1068 sharing. The gain of SALS is therefore the gain of fine-tuning that rule,
1069 and we cannot say whether a better optimiser or a richer policy class
1070 would find larger gains. The controller uses the simulated state of the
1071 crop and soil (thermal time, canopy interception, soil water) without
1072 measurement error, and the forecast errors we tested apply to weather
1073 only.
- 1074 • *Economics and deployment.* The economic analysis (Section 3.9) uses
1075 assumed prices and costs, no grid, curtailment, storage or subsidy
1076 model, and does not value food security; its break-even prices move
1077 with the capex and the discount rate (Fig. 7). The value of midday
1078 electricity, curtailment, storage, structural cost and farmer acceptance
1079 (Trommsdorff et al., 2021) will determine the realisable share. The av-
1080 erage row pitch at the chosen densities (about 13 m for 2.2-m modules)
1081 leaves room for machinery, but the densest designs (GCR up to 0.5)
1082 have a pitch of only 4.4 m.

5. Conclusions

Light sharing, the daily choice between maximum electricity and letting light through to the crop, is the main lever on the electricity of tracker-based agrivoltaics at a given yield-retention floor, although it pays only at high electricity prices: below about 80 USD MWh⁻¹ static arrays earn more. In a differentiable simulator of trackers, PV output, crop light, microclimate and crop growth, evaluated on held-out crop–site pairs (92 on the main split, 76 distinct pairs pooled over three splits for the controller and design comparisons) with the array density of each site chosen from its own history, light-sharing rules allowed 2.1 to 2.9 times the module density of static arrays at a mean yield retention of 90%. They raised the land-equivalent ratio from 1.07 to 1.18–1.24, and a rule that follows crop phenology and needs no weather forecast was the best of them. The simulated shade response is conservative against the field trials we could use for rice, wheat and potato, and calibrating it raised the gains without changing the ranking of methods; for maize and soybean, which supply 53% of the upscaled electricity, the meta-analysis indicates larger losses than the simulator reproduces whereas one maize trial indicates smaller ones, so the gains for these crops are uncertain and possibly overestimated. The crop’s response to shade, the least certain input, was learned from 428 field observations: a hierarchical model predicted a held-out study of the staple crop types no better than proportional loss (error 0.22), which shows the size of the between-study spread, and gradient descent through the simulator turned it into an ensemble of plausible responses. A learned controller exceeded the written phenology rule by 0.044 LER and a rule with a window tuned on the training seasons by 0.018 (95% interval 0.012–0.024), by following a smooth phenology-dependent schedule that a rule can express; training it on ensembles of shade responses or giving it a farm-measurable biomass signal did not help, and with the wider ensembles it did not retain its margin over the written rule in our training budget (one seed, 21 pairs). The decision that mattered was the design. A density chosen under the mean response left about 70% of crop–site pairs at the floor when another plausible response held (oracle 85–87%); choosing the density so that the floor holds in 75% of the plausible responses recovered about half of the gap at a cost of 0.04 LER, but a floor raised by 0.02 under the mean response performed alike at equal compliance (differences of -0.006 to $+0.024$ LER with intervals containing zero), and no rule was safe outside the ensemble. The ensemble is therefore useful to size a safety

margin, not to replace it. Under illustrative costs, light sharing breaks even with the open field at 70 to 92 USD MWh⁻¹ (73 for the learned controller, 59 for static arrays); a conventional plant replacing the crop is more profitable above its break-even of 48 USD MWh⁻¹, so agrivoltaics depends on a premium, land-use rules or a value for food production that markets do not give. The yield floor is met on average but not in every season or region, and the electricity potential of 5% of staple cropland (about 14 PWh yr⁻¹) requires about 10 TWp of modules. Future work should validate the crop response to shade under trackers with more field data, notably for maize and legumes where the meta-analysis and the trials disagree, extend the crop model to grain-number formation and canopy energy balance, add costs and grid constraints, and pursue controllers that guarantee the yield-retention floor per site and per season.

CRedit authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

All data used are public: SPAM 2010 v2r0 harvested areas (Yu et al., 2020), the GGCM Phase 3 crop calendar (Jägermeyr et al., 2021), NASA POWER daily weather (Sparks, 2018) the SIMPLE crop parameters (Zhao et al., 2019) and the shade-response meta-analysis data of Laub et al. (2022) (Zenodo 10.5281/zenodo.5716091, CC BY 4.0, copied into the repository). The code of the simulator, of SALS and of all experiments, together with the site list, is available from the author’s public repository (<https://github.com/vinhqdang/Artificial-Intelligence-in-Agriculture>).

1150 **Funding**

1151 This research did not receive any specific grant from funding agencies in
1152 the public, commercial, or not-for-profit sectors.

1153 **Appendix A. Statistical comparison**

1154 **References**

- 1155 Adeh, E.H., Good, S.P., Calaf, M., Higgins, C.W., 2019. Solar pv power
1156 potential is greatest over croplands. *Scientific Reports* 9, 11442. doi:10.
1157 1038/s41598-019-47803-3.
- 1158 Allen, R.G., Pereira, L.S., Raes, D., Smith, M., 1998. Crop evapotranspira-
1159 tion: Guidelines for computing crop water requirements. *FAO Irrigation
1160 and Drainage Paper* 56. Food and Agriculture Organization of the United
1161 Nations. Rome.
- 1162 Amaducci, S., Yin, X., Colauzzi, M., 2018. Agrivoltaic systems to optimise
1163 land use for electric energy production. *Applied Energy* 220, 545–561.
1164 doi:10.1016/j.apenergy.2018.03.081.
- 1165 Andrade, F.H., Vega, C., Uhart, S., Cirilo, A., Cantarero, M., Valentinuz, O.,
1166 1999. Kernel number determination in maize. *Crop Science* 39, 453–459.
1167 doi:10.2135/cropsci1999.0011183X0039000200026x.
- 1168 Barron-Gafford, G.A., Pavao-Zuckerman, M.A., Minor, R.L., Sutter, L.F.,
1169 Barnett-Moreno, I., Blackett, D.T., Thompson, M., Dimond, K., Gerlak,
1170 A.K., Nabhan, G.P., Macknick, J.E., 2019. Agrivoltaics provide mutual
1171 benefits across the food–energy–water nexus in drylands. *Nature Sustain-
1172 ability* 2, 848–855. doi:10.1038/s41893-019-0364-5.
- 1173 Calafiore, G.C., Campi, M.C., 2006. The scenario approach to robust control
1174 design. *IEEE Transactions on Automatic Control* 51, 742–753. doi:10.
1175 1109/TAC.2006.875041.
- 1176 Campana, P.E., Stridh, B., Amaducci, S., Colauzzi, M., 2021. Optimisation
1177 of vertically mounted agrivoltaic systems. *Journal of Cleaner Production*
1178 325, 129091. doi:10.1016/j.jclepro.2021.129091.

1179 Campi, M.C., Garatti, S., 2008. The exact feasibility of randomized solutions
1180 of uncertain convex programs. *SIAM Journal on Optimization* 19, 1211–
1181 1230. doi:10.1137/07069821X.

1182 Collares-Pereira, M., Rabl, A., 1979. The average distribution of solar
1183 radiation-correlations between diffuse and hemispherical and between daily
1184 and hourly insolation values. *Solar Energy* 22, 155–164. doi:10.1016/
1185 0038-092x(79)90100-2.

1186 DIN, 2021. DIN SPEC 91434:2021-05: Agri-photovoltaic systems – require-
1187 ments for primary agricultural use. Beuth Verlag, Berlin.

1188 Dinesh, H., Pearce, J.M., 2016. The potential of agrivoltaic systems. *Renew-
1189 able and Sustainable Energy Reviews* 54, 299–308. doi:10.1016/j.rser.
1190 2015.10.024.

1191 Dupraz, C., 2023. Assessment of the ground coverage ratio of agrivoltaic
1192 systems as a proxy for potential crop productivity. *Agroforestry Systems*
1193 98, 2679–2696. doi:10.1007/s10457-023-00906-3.

1194 Dupraz, C., Marrou, H., Talbot, G., Dufour, L., Nogier, A., Ferard, Y., 2011.
1195 Combining solar photovoltaic panels and food crops for optimising land
1196 use: Towards new agrivoltaic schemes. *Renewable Energy* 36, 2725–2732.
1197 doi:10.1016/j.renene.2011.03.005.

1198 Elamri, Y., Cheviron, B., Lopez, J.M., Dejean, C., Belaud, G., 2018. Water
1199 budget and crop modelling for agrivoltaic systems: Application to irrigated
1200 lettuces. *Agricultural Water Management* 208, 440–453. doi:10.1016/j.
1201 agwat.2018.07.001.

1202 Ember, 2024. Global Electricity Review 2024. Technical Report. Ember.
1203 London.

1204 European Commission, Joint Research Centre, 2022. Photovoltaic geo-
1205 graphical information system (PVGIS), version 5.2. [https://re.jrc.ec.
1206 europa.eu/pvg_tools/](https://re.jrc.ec.europa.eu/pvg_tools/).

1207 F. Holmgren, W., W. Hansen, C., A. Mikofski, M., 2018. pvlib python: a
1208 python package for modeling solar energy systems. *Journal of Open Source
1209 Software* 3, 884. doi:10.21105/joss.00884.

1210 Fischer, R.A., 1985. Number of kernels in wheat crops and the influence of
1211 solar radiation and temperature. *The Journal of Agricultural Science* 105,
1212 447–461. doi:10.1017/S0021859600056495.

1213 Gautron, R., Maillard, O.A., Preux, P., Corbeels, M., Sabbadin, R., 2022.
1214 Reinforcement learning for crop management support: Review, prospects
1215 and challenges. *Computers and Electronics in Agriculture* 200, 107182.
1216 doi:10.1016/j.compag.2022.107182.

1217 Gonocruz, R.A., Nakamura, R., Yoshino, K., Homma, M., Doi, T., Yoshida,
1218 Y., Tani, A., 2021. Analysis of the rice yield under an agrivoltaic
1219 system: A case study in japan. *Environments* 8, 65. doi:10.3390/
1220 environments8070065.

1221 Hassanpour Adeg, E., Selker, J.S., Higgins, C.W., 2018. Remarkable agri-
1222 voltaic influence on soil moisture, micrometeorology and water-use effi-
1223 ciency. *PLOS ONE* 13, e0203256. doi:10.1371/journal.pone.0203256.

1224 IPCC, 2021. Climate Change 2021: The Physical Science Basis. Contribution
1225 of Working Group I to the Sixth Assessment Report of the Intergovern-
1226 mental Panel on Climate Change. Cambridge University Press, Cambridge.
1227 doi:10.1017/9781009157896.

1228 IRENA, 2025. Renewable Capacity Statistics 2025. Technical Report. Inter-
1229 national Renewable Energy Agency. Abu Dhabi.

1230 Jägermeyr, J., Müller, C., Ruane, A.C., Elliott, J., Balkovic, J., Castillo, O.,
1231 Faye, B., Foster, I., Folberth, C., Franke, J.A., Fuchs, K., Guarin, J.R.,
1232 Heinke, J., Hoogenboom, G., Iizumi, T., Jain, A.K., Kelly, D., Khabarov,
1233 N., Lange, S., Lin, T.S., Liu, W., Mialyk, O., Minoli, S., Moyer, E.J.,
1234 Okada, M., Phillips, M., Porter, C., Rabin, S.S., Scheer, C., Schneider,
1235 J.M., Schyns, J.F., Skalsky, R., Smerald, A., Stella, T., Stephens, H.,
1236 Webber, H., Zabel, F., Rosenzweig, C., 2021. Climate impacts on global
1237 agriculture emerge earlier in new generation of climate and crop models.
1238 *Nature Food* 2, 873–885. doi:10.1038/s43016-021-00400-y.

1239 Karniadakis, G.E., Kevrekidis, I.G., Lu, L., Perdikaris, P., Wang, S., Yang,
1240 L., 2021. Physics-informed machine learning. *Nature Reviews Physics* 3,
1241 422–440. doi:10.1038/s42254-021-00314-5.

1242 Kingma, D.P., Ba, J., 2015. Adam: A method for stochastic optimization,
1243 in: International Conference on Learning Representations.

1244 Laub, M., Pataczek, L., Feuerbacher, A., Zikeli, S., Högy, P., 2022. Contrast-
1245 ing yield responses at varying levels of shade suggest different suitability
1246 of crops for dual land-use systems: a meta-analysis. *Agronomy for Sus-*
1247 *tainable Development* 42, 51. doi:10.1007/s13593-022-00783-7.

1248 Liu, B.Y., Jordan, R.C., 1960. The interrelationship and characteristic dis-
1249 tribution of direct, diffuse and total solar radiation. *Solar Energy* 4, 1–19.
1250 doi:10.1016/0038-092x(60)90062-1.

1251 Long, S.P., Ainsworth, E.A., Leakey, A.D.B., Nösberger, J., Ort, D.R., 2006.
1252 Food for thought: Lower-than-expected crop yield stimulation with ris-
1253 ing CO_2 concentrations. *Science* 312, 1918–1921. doi:10.1126/science.
1254 1114722.

1255 Lorenzo, E., Narvarte, L., Muñoz, J., 2011. Tracking and back-tracking.
1256 *Progress in Photovoltaics: Research and Applications* 19, 747–753. doi:10.
1257 1002/pip.1085.

1258 Mamun, M.A.A., Dargusch, P., Wadley, D., Zulkarnain, N.A., Aziz, A.A.,
1259 2022. A review of research on agrivoltaic systems. *Renewable and Sustain-*
1260 *able Energy Reviews* 161, 112351. doi:10.1016/j.rser.2022.112351.

1261 Marchau, V.A.W.J., Walker, W.E., Bloemen, P.J.T.M., Popper, S.W. (Eds.),
1262 2019. *Decision Making under Deep Uncertainty: From Theory to Practice*.
1263 Springer. doi:10.1007/978-3-030-05252-2.

1264 Marrou, H., Dufour, L., Wery, J., 2013a. How does a shelter of solar panels
1265 influence water flows in a soil–crop system? *European Journal of Agron-*
1266 *omy* 50, 38–51. doi:10.1016/j.eja.2013.05.004.

1267 Marrou, H., Guillioni, L., Dufour, L., Dupraz, C., Wery, J., 2013b. Microcli-
1268 mate under agrivoltaic systems: Is crop growth rate affected in the partial
1269 shade of solar panels? *Agricultural and Forest Meteorology* 177, 117–132.
1270 doi:10.1016/j.agrformet.2013.04.012.

1271 Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen,
1272 T., Lin, Z., Gimelshein, N., Antiga, L., et al., 2019. PyTorch: An impera-
1273 tive style, high-performance deep learning library, in: *Advances in Neural*
1274 *Information Processing Systems*, pp. 8024–8035.

1275 Ramos-Fuentes, I.A., Elamri, Y., Cheviron, B., Dejean, C., Belaud, G.,
1276 Fumey, D., 2023. Effects of shade and deficit irrigation on maize growth
1277 and development in fixed and dynamic agrivoltaic systems. *Agricultural*
1278 *Water Management* 280, 108187. doi:10.1016/j.agwat.2023.108187.

1279 Reichstein, M., Camps-Valls, G., Stevens, B., Jung, M., Denzler, J., Car-
1280 valhais, N., Prabhat, 2019. Deep learning and process understanding for
1281 data-driven Earth system science. *Nature* 566, 195–204. doi:10.1038/
1282 s41586-019-0912-1.

1283 Sekiyama, T., Nagashima, A., 2019. Solar sharing for both food and
1284 clean energy production: Performance of agrivoltaic systems for corn,
1285 a typical shade-intolerant crop. *Environments* 6, 65. doi:10.3390/
1286 environments6060065.

1287 Shen, C., Appling, A.P., Gentine, P., Bandai, T., Gupta, H., Tartakovsky, A.,
1288 Baity-Jesi, M., Fenicia, F., Kifer, D., Li, L., Liu, X., Ren, W., Zheng, Y.,
1289 Harman, C.J., Clark, M., Farthing, M., Feng, D., Kumar, P., Aboelyazeed,
1290 D., Rahmani, F., Song, Y., Beck, H.E., Bindas, T., Dwivedi, D., Fang,
1291 K., Höge, M., Rackauckas, C., Mohanty, B., Roy, T., Xu, C., Lawson,
1292 K., 2023. Differentiable modelling to unify machine learning and physical
1293 models for geosciences. *Nature Reviews Earth & Environment* 4, 552–567.
1294 doi:10.1038/s43017-023-00450-9.

1295 Siebert, S., Ewert, F., Eyshi Rezaei, E., Kage, H., Graß, R., 2014. Impact
1296 of heat stress on crop yield—on the importance of considering canopy
1297 temperature. *Environmental Research Letters* 9, 044012. doi:10.1088/
1298 1748-9326/9/4/044012.

1299 Sparks, A., 2018. nasapower: A nasa power global meteorology, surface solar
1300 energy and climatology data client for r. *Journal of Open Source Software*
1301 3, 1035. doi:10.21105/joss.01035.

1302 Trommsdorff, M., Kang, J., Reise, C., Schindele, S., Bopp, G., Ehmann, A.,
1303 Weselek, A., Högy, P., Obergfell, T., 2021. Combining food and energy
1304 production: Design of an agrivoltaic system applied in arable and vegetable
1305 farming in germany. *Renewable and Sustainable Energy Reviews* 140,
1306 110694. doi:10.1016/j.rser.2020.110694.

- 1307 Valle, B., Simonneau, T., Sourd, F., Pechier, P., Hamard, P., Frisson, T.,
1308 Ryckewaert, M., Christophe, A., 2017. Increasing the total productivity of
1309 a land by combining mobile photovoltaic panels and food crops. *Applied*
1310 *Energy* 206, 1495–1507. doi:10.1016/j.apenergy.2017.09.113.
- 1311 Waha, K., Dietrich, J.P., Portmann, F.T., Siebert, S., Thornton, P.K., Bon-
1312 deau, A., Herrero, M., 2020. Multiple cropping systems of the world
1313 and the potential for increasing cropping intensity. *Global Environmental*
1314 *Change* 64, 102131. doi:10.1016/j.gloenvcha.2020.102131.
- 1315 Webber, H., White, J.W., Kimball, B.A., Ewert, F., Asseng, S., Eyshi Rezaei,
1316 E., Pinter, P.J., Hatfield, J.L., Reynolds, M.P., Ababaei, B., Bindi, M.,
1317 Doltra, J., Ferrise, R., Kage, H., Kassie, B.T., Kersebaum, K.C., Luig,
1318 A., Olesen, J.E., Semenov, M.A., Stratonovitch, P., Ratjen, A.M., LaM-
1319 orte, R.L., Leavitt, S.W., Hunsaker, D.J., Wall, G.W., Martre, P., 2018.
1320 Physical robustness of canopy temperature models for crop heat stress
1321 simulation across environments and production conditions. *Field Crops*
1322 *Research* 216, 75–88. doi:10.1016/j.fcr.2017.11.005.
- 1323 Weselek, A., Bauerle, A., Hartung, J., Zikeli, S., Lewandowski, I., Högy, P.,
1324 2021. Agrivoltaic system impacts on microclimate and yield of different
1325 crops within an organic crop rotation in a temperate climate. *Agronomy*
1326 *for Sustainable Development* 41, 59. doi:10.1007/s13593-021-00714-y.
- 1327 Weselek, A., Ehmann, A., Zikeli, S., Lewandowski, I., Schindele, S., Högy,
1328 P., 2019. Agrophotovoltaic systems: applications, challenges, and op-
1329 portunities. a review. *Agronomy for Sustainable Development* 39, 35.
1330 doi:10.1007/s13593-019-0581-3.
- 1331 Willockx, B., Lavaert, C., Cappelle, J., 2023. Performance evaluation of
1332 vertical bifacial and single-axis tracked agrivoltaic systems on arable land.
1333 *Renewable Energy* 217, 119181. doi:10.1016/j.renene.2023.119181.
- 1334 Woli, P., Jones, J.W., Ingram, K.T., Fraisse, C.W., 2012. Agricultural
1335 reference index for drought (arid). *Agronomy Journal* 104, 287–300.
1336 doi:10.2134/agronj2011.0286.
- 1337 Yu, Q., You, L., Wood-Sichra, U., Ru, Y., Joglekar, A.K.B., Fritz, S., Xiong,
1338 W., Lu, M., Wu, W., Yang, P., 2020. A cultivated planet in 2010 – part 2:

- 1339 The global gridded agricultural-production maps. *Earth System Science*
1340 *Data* 12, 3545–3572. doi:10.5194/essd-12-3545-2020.
- 1341 Zhang, L., Geng, X., Ding, H., Cao, K., Wang, L., Chen, H., Deng, L., Wu,
1342 C., Xiao, M., Bao, E., 2026. Multi-objective optimization of photovoltaic
1343 array layouts on farmland via a hybrid algorithm framework for enhanced
1344 agricultural production and energy generation. *Artificial Intelligence in*
1345 *Agriculture* doi:10.1016/j.aiaa.2026.09.002.
- 1346 Zhao, C., Liu, B., Piao, S., Wang, X., Lobell, D.B., Huang, Y., Huang, M.,
1347 Yao, Y., Bassu, S., Ciais, P., Durand, J.L., Elliott, J., Ewert, F., Janssens,
1348 I.A., Li, T., Lin, E., Liu, Q., Martre, P., Müller, C., Peng, S., Peñuelas,
1349 J., Ruane, A.C., Wallach, D., Wang, T., Wu, D., Liu, Z., Zhu, Y., Zhu,
1350 Z., Asseng, S., 2017. Temperature increase reduces global yields of major
1351 crops in four independent estimates. *Proceedings of the National Academy*
1352 *of Sciences* 114, 9326–9331. doi:10.1073/pnas.1701762114.
- 1353 Zhao, C., Liu, B., Xiao, L., Hoogenboom, G., Boote, K.J., Kassie, B.T.,
1354 Pavan, W., Shelia, V., Kim, K.S., Hernandez-Ochoa, I.M., Wallach, D.,
1355 Porter, C.H., Stockle, C.O., Zhu, Y., Asseng, S., 2019. A simple crop
1356 model. *European Journal of Agronomy* 104, 97–106. doi:10.1016/j.eja.
1357 2019.01.009.

Table 2: Held-out performance (92 test sites, seasons 2015–2019, canopy cooling $\kappa = 1^\circ\text{C}$). The density (GCR) of each site is the largest value on the grid 0.05, 0.075, ..., 0.5 at which the method’s mean retention in the site’s 2001–2014 seasons is ≥ 0.9 . E : electricity ($\text{MWh ha}^{-1} \text{ yr}^{-1}$); e : electricity relative to a conventional PV plant; r : mean yield retention; r_{prod} : production-weighted retention; C_s and C_{site} : share of seasons with $r \geq 0.9$ and share of sites whose test-season mean $r \geq 0.9$ (%); $\text{LER} = r + e$.

Method	GCR	E	e	r	r_{prod}	C_s	C_{site}	LER
<i>Baseline</i>								
AV static	0.06	217	0.16	0.917	0.914	93	96	1.07
Seasonal sharing	0.17	397	0.28	0.904	0.902	69	72	1.19
Phenology rule	0.17	465	0.33	0.904	0.902	69	75	1.24
Stress rule	0.13	387	0.28	0.907	0.907	76	82	1.18
SALS (ours)	0.17	510	0.36	0.905	0.902	70	74	1.27
<i>+1.5 °C</i>								
AV static	0.06	221	0.16	0.916	0.914	93	92	1.08
Seasonal sharing	0.16	393	0.28	0.905	0.903	72	74	1.19
Phenology rule	0.17	478	0.34	0.905	0.902	68	75	1.25
Stress rule	0.12	377	0.27	0.908	0.908	73	77	1.18
SALS (ours)	0.17	518	0.37	0.906	0.903	68	75	1.28
<i>+2 °C</i>								
AV static	0.06	223	0.16	0.916	0.914	92	93	1.08
Seasonal sharing	0.17	394	0.28	0.905	0.904	72	76	1.19
Phenology rule	0.17	483	0.35	0.905	0.903	69	74	1.25
Stress rule	0.12	370	0.26	0.909	0.909	75	78	1.17
SALS (ours)	0.17	521	0.38	0.907	0.904	69	75	1.28
<i>+3 °C</i>								
AV static	0.06	231	0.17	0.915	0.914	88	89	1.08
Seasonal sharing	0.17	397	0.29	0.906	0.904	69	75	1.19
Phenology rule	0.18	499	0.36	0.905	0.902	67	72	1.26
Stress rule	0.11	366	0.26	0.908	0.908	74	77	1.17
SALS (ours)	0.18	540	0.39	0.907	0.904	68	75	1.30

Table 3: LER of the controllers at different yield-retention floors ρ (main split; the historical design step is repeated for each floor).

Method	Baseline climate				+3 °C			
	0.80	0.85	0.90	0.95	0.80	0.85	0.90	0.95
AV static	1.159	1.115	1.072	1.011	1.174	1.128	1.083	1.013
Stress rule	1.380	1.282	1.182	1.077	1.364	1.269	1.172	1.071
Phenology rule	1.467	1.356	1.236	1.108	1.513	1.395	1.265	1.122
SALS (trained at $\rho = 0.9$)	1.505	1.392	1.270	1.141	1.553	1.432	1.297	1.154
SALS conditioned on ρ	1.504	1.391	1.269	1.141	1.553	1.431	1.296	1.154

Table 4: Sensitivity of the controller comparison to the rotation limit of the trackers and to the harvest-index penalty (LER; main split, every sixth held-out pair, 21 pairs, main shade response, history-based design on a GCR grid with step 0.05). SALS was trained for the main setting. Fixed arrays (no tracking; flat and tilted by 30° from the horizontal about the north–south axis) reach LER 1.048 and 1.043 in the same sample, against 1.053 for static trackers.

Setting	Static tracker	Phenology rule	Tuned rule	SALS
Main (limit 60°, window 0.45–0.65)	1.053	1.170	1.180	1.189
Rotation limit 45°	1.057	1.138	1.144	1.144
Rotation limit 80°	1.053	1.206	1.219	1.224
Window 0.30–0.50	1.053	1.169	1.176	1.185
Window 0.60–0.80	1.053	1.171	1.181	1.182
No harvest-index penalty	1.109	1.328	1.347	1.353

Table 5: Rules and the learned controller pooled over the three spatial splits (76 distinct crop-site pairs, every third held-out pair; baseline climate; main shade response; history-based design on a GCR grid with step 0.05, coarser than the 0.025 grid of the main analysis). The tuned phenology rule shares light while thermal time lies between 0.15 and 1.0 of the season requirement (window chosen on training-block seasons of split 0; the written rule uses 0.05 to 1.0). The optimised ramp schedule has five parameters (start 0.17, end 0.93, ramp widths 0.02, level 0.99) found by CMA-ES on the same training seasons.

Method	GCR	E/E_{PV}	Retention	Seasons ≥ 0.9 (%)	Pairs ≥ 0.9 (%)	LER
AV static	0.059	0.153	0.922	95	96	1.076
Seasonal sharing	0.151	0.249	0.915	84	91	1.164
Stress rule	0.118	0.255	0.918	83	88	1.173
Phenology rule	0.154	0.292	0.916	85	91	1.207
Tuned phenology rule	0.161	0.317	0.916	84	89	1.233
Optimised ramp schedule	0.155	0.322	0.916	83	87	1.238
SALS	0.161	0.335	0.916	83	87	1.251

Table 6: Simulated relative yield at the shading of published field trials (static arrays, training-block sites). Observed values: rice, single trial; wheat and potato, the worse of the two seasons (the better season gave +2.7% and +11%); maize, mean of the three seasons of the dynamic system (DAV) and single values for the other two systems. The same trials enter the calibration of Section 2.6, so the comparison is not independent.

Trial	Shading	Observed	Conservative	Calibrated
Rice, Gonocruz et al. (2021)	0.25	0.84	0.77	0.79
Wheat, Weselek et al. (2021)	0.30	0.81	0.66	0.74
Potato, Weselek et al. (2021)	0.30	0.82	0.72	0.72
Maize, Ramos-Fuentes et al. (2023) (DAV)	0.35	0.83	0.70	0.70
Maize, Ramos-Fuentes et al. (2023) (half)	0.30	0.91	0.75	0.75
Maize, Ramos-Fuentes et al. (2023) (full)	0.50	0.70	0.54	0.54

Table 7: Sensitivity of the deployable light-sharing results to the crop shade response (rule controllers, baseline climate, floor $\rho = 0.9$, split 0, every fourth held-out site, 22 sites). Conservative: proportional response (main analysis); calibrated: radiation-use compensation fitted to Laub et al. (2022); field-fitted: fitted to the field trials, maize harvest index unaffected.

Shade response	Method	GCR	E/E_{PV}	Retention	LER
conservative	AV static	0.06	0.15	0.92	1.07
	Phenology rule	0.17	0.33	0.90	1.23
	Seasonal sharing	0.16	0.28	0.90	1.18
	Stress rule	0.11	0.24	0.91	1.15
calibrated	AV static	0.07	0.18	0.91	1.09
	Phenology rule	0.19	0.37	0.90	1.28
	Seasonal sharing	0.19	0.32	0.90	1.22
	Stress rule	0.13	0.29	0.91	1.20
field-fitted	AV static	0.16	0.41	0.91	1.32
	Phenology rule	0.37	0.70	0.92	1.62
	Seasonal sharing	0.37	0.61	0.92	1.53
	Stress rule	0.29	0.63	0.90	1.53

Table 8: Predicting relative yield under shade from 428 observations of 58 studies (Laub et al., 2022): leave-one-study-out cross-validation (every study held out once; observations with $s > 0$). Covariates: irradiation, climate zone, shade type, intercropping.

Model	RMSE	MAE	Median AE	Mean MAE per study
Proportional loss, $y = 1 - s$	0.430	0.305	0.210	0.224
One curve for all crops	0.372	0.265	0.193	0.208
Curve per crop type, no pooling	0.313	0.208	0.141	0.165
Curve per crop type, partial pooling	0.312	0.210	0.150	0.162
Partial pooling + covariates	0.311	0.206	0.148	0.161
Gradient-boosted trees	0.321	0.219	0.162	0.184

Table 9: Leave-one-study-out prediction error (RMSE of relative yield, observations with $s > 0$) by crop type: proportional loss, hierarchical curve with partial pooling, and gradient-boosted trees. The last columns give the difference hierarchical minus proportional with a 95% cluster-bootstrap interval over studies (negative: hierarchical better).

Group	Obs./studies	Proportional	Hierarchical	Boosted t
C3 cereals	30/10	0.214	0.187	0.205
Maize	40/10	0.225	0.235	0.248
Grain legumes	58/14	0.215	0.225	0.234
Tubers/root crops	10/2	0.250	0.254	0.257
Four crop types used by the simulator	138/30	0.220	0.222	0.234
Other types (berries, forages, fruits, vegetables)	201/29	0.528	0.361	0.369
All types	339/58	0.430	0.312	0.321

Table 10: Relative yield at 20% and 40% shading: data-driven model (median and 80% bootstrap interval over studies; crop-type curve, covariates at their reference values: mean irradiation, temperate zone, no solar-panel shade, no intercropping) and simulator under the two calibrated ensembles (mean of members; E1 fitted to hand-read curves, E2 fitted to the data-driven curves). The data-driven curve of rice is that of C3 cereals (no rice in the data set). Observations and studies: C3 cereals 41/10, maize 51/10, grain legumes 76/14, tubers 12/2.

Crop	Shading	Data-driven	Simulator, E1	Simulator, E2	Proportional
Wheat	20%	0.93 [0.88, 0.96]	0.88	0.92	0.80
Wheat	40%	0.72 [0.59, 0.78]	0.70	0.77	0.60
Rice	20%	0.93 [0.88, 0.96]	0.85	0.90	0.80
Rice	40%	0.72 [0.59, 0.78]	0.66	0.73	0.60
Maize	20%	0.73 [0.58, 0.86]	0.86	0.90	0.80
Maize	40%	0.42 [0.27, 0.59]	0.69	0.75	0.60
Soybean	20%	0.71 [0.65, 0.79]	0.77	0.74	0.80
Soybean	40%	0.40 [0.34, 0.53]	0.53	0.48	0.60
Potato	20%	0.93 [0.92, 0.97]	0.88	0.92	0.80
Potato	40%	0.71 [0.67, 0.83]	0.73	0.79	0.60

Table 11: Shade-response parameters of ensemble E2, calibrated by gradient descent through the simulator to the data-driven curves and the field trials (12 randomised fits; mean $\pm$ s.d.): radiation-use compensation c and multiplier h on the flowering-window harvest-index penalty (1 = assumed value).

Crop	c	h multiplier
Wheat	0.91 ± 0.31	0.74 ± 0.53
Rice	0.61 ± 0.25	0.83 ± 0.33
Maize	0.14 ± 0.11	0.16 ± 0.13
Soybean	0.05 ± 0.09	1.66 ± 0.38
Potato	0.62 ± 0.20	0.90 ± 0.50

Table 12: Land-equivalent ratio of rules and learned controllers under four true shade responses, with the array density designed under the true response (split 0, baseline climate, floor $\rho = 0.9$, every sixth held-out pair, 21 pairs, coarse GCR grid, one training seed; season means, so differences differ slightly from the pair-level intervals in the text). SALS is trained on the proportional response; robust controllers are trained on random members of ensemble E1, E2 or E3 (adaptive: with the measured biomass ratio). The proportional and harsh responses lie outside the ensembles.

Method	Proportional	Calibrated	Field-fitted	Harsh
AV static	1.052	1.059	1.254	1.037
Seasonal sharing	1.133	1.156	1.385	1.097
Stress rule	1.129	1.160	1.406	1.086
Feedback rule	1.139	1.164	1.392	1.089
Phenology rule	1.164	1.190	1.456	1.122
Tuned phenology rule	1.170	1.199	1.472	1.129
Optimised ramp schedule	1.170	1.199	1.477	1.122
SALS	1.178	1.209	1.489	1.132
SALS-robust, E1	1.179	1.204	1.491	1.127
SALS-adaptive, E1	1.176	1.202	1.495	1.124
SALS-robust, E2	1.165	1.192	1.483	1.126
SALS-robust, E3	1.163	1.191	1.481	1.121
SALS-adaptive, E3	1.164	1.191	1.475	1.118

Table 13: Array design under shade-response uncertainty, phenology rule, three spatial splits pooled (76 distinct crop-site pairs, every third held-out pair). Truth: each of five members of the ensemble in turn; the mean response and the member-based rules use the other four members (leave-one-out, so the truth never enters the design). E2: ensemble calibrated to the learned curves and field trials; E3: same procedure with target errors of 0.2 on the learned curves. Pairs: share with mean retention ≥ 0.9 ; shortfall: mean of $\max(0, 0.9 - \text{retention})$ per pair. The LER values exceed those of the main analysis because E2 and E3 are milder than the proportional response and allow denser arrays; the margin grid (0.92, 0.94, 0.96) was chosen after a first analysis of E2 and is not pre-specified.

Design	E2			E3		
	Pairs ≥ 0.9 (%)	Shortfall	LER	Pairs ≥ 0.9 (%)	Shortfall	LER
oracle (true response)	87	0.0025	1.486	85	0.0029	1.486
mean response	70	0.0062	1.493	71	0.0093	1.493
floor met in 50% of other members	65	0.0084	1.510	67	0.0125	1.510
floor met in 75% of other members	78	0.0045	1.462	79	0.0083	1.462
floor met in all other members	89	0.0019	1.412	88	0.0032	1.412
floor 0.92 under mean response	88	0.0022	1.426	85	0.0051	1.426
floor 0.94 under mean response	97	0.0007	1.345	92	0.0027	1.345
floor 0.96 under mean response	99	0.0003	1.251	96	0.0015	1.251

Table 14: Design rules outside the ensemble: share of crop-site pairs (%) with mean retention ≥ 0.9 under four fixed responses (P proportional, C calibrated to the meta-analysis, F field-fitted, H harsh), phenology rule, three splits pooled (76 distinct pairs). Designs use the whole ensemble E2 or E3 (five members and their mean); the oracle uses the true response. LER in the last column is the mean over the four responses.

Design	E2				E3				LER	
	P	C	F	H	P	C	F	H	E2	E3
oracle (true response)	91	89	87	84	91	89	87	84	1.283	1.283
mean response	13	13	68	12	13	17	79	7	1.435	1.448
floor met in 75% of members	16	16	80	12	22	25	93	13	1.384	1.397
floor met in all members	17	17	80	12	29	32	93	16	1.348	1.336
floor 0.92 under mean response	17	20	86	12	29	33	91	20	1.376	1.399
floor 0.94 under mean response	45	47	97	34	45	45	97	29	1.303	1.345
floor 0.96 under mean response	63	78	99	54	58	63	99	51	1.220	1.264

Table 15: Assumed economic parameters (illustrative, not data). Ranges used in the sensitivity analysis are in parentheses.

Parameter	Value
Electricity price	60 USD MWh ⁻¹ (30–120)
Capex, AV single-axis tracker plant	1.0 USD Wp ⁻¹ (0.7–1.5)
Capex, conventional tracker plant	0.75 USD Wp ⁻¹
Discount rate, lifetime, O&M	6% (4–8%), 25 years, 1.5% of capex per year
Installed capacity	2.1 MWp per hectare of module area share g (0.21 kWp m ⁻²)
Crop price per tonne of dry matter	wheat 290, rice 400, maize 250, soybean 520, potato 1000 USD (0.

Table 16: Illustrative annual economics per hectare (76 crop–site pairs, three splits, main shade response, baseline climate; parameters in Table 15). Net value is relative to the open field (crop costs unchanged) in kUSD ha⁻¹ yr⁻¹ at three electricity prices; break-even is the price at which the net value is zero. The conventional plant replaces the crop.

Method	E (MWh)	Capex annuity	Crop loss	Net at 60	Net at 90	Net at 120
AV static	200	11.5	0.27	0.3	6.3	12.3
Seasonal sharing	326	29.6	0.31	-10.3	-0.5	9.2
Stress rule	336	23.2	0.30	-3.3	6.8	16.9
Phenology rule	382	30.1	0.30	-7.5	4.0	15.4
Tuned phenology rule	415	31.6	0.31	-7.0	5.5	17.9
SALS	438	31.6	0.31	-5.6	7.6	20.7
Conventional PV plant	1306	58.7	3.60	16.0	55.2	94.4

Table 17: One-at-a-time sensitivity of the net value (kUSD ha⁻¹ yr⁻¹, relative to the open field) at 90 USD MWh⁻¹, densities designed for the floor as in the main analysis. In the capex rows the capex of the conventional plant is set to 0.75 times that of the agrivoltaic plant.

Parameter	Value	Static arrays	Tuned rule	SALS	Conventional plant
Base		6.3	5.5	7.6	55.2
AV capex (USD/Wp)	0.7	9.7	14.9	17.1	72.8
AV capex (USD/Wp)	1.5	0.6	-10.3	-8.2	25.8
Discount rate	4%	8.0	10.3	12.4	64.1
Discount rate	8%	4.4	0.2	2.4	45.4
Crop prices	0.5×	6.4	5.6	7.7	57.0
Crop prices	2.0×	6.0	5.2	7.3	51.6
Electricity price	60	0.3	-7.0	-5.6	16.0
Electricity price	120	12.3	17.9	20.7	94.4

Table A.18: Paired comparison of SALS with each baseline on the 92 held-out sites of the main split (unit of analysis: site mean over test seasons). ΔLER : mean difference (SALS minus baseline) with cluster-bootstrap 95% CI; p : two-sided Wilcoxon signed-rank test; % better: share of sites where SALS has the higher LER; Δr : difference in mean yield retention with its Wilcoxon p -value.

Climate	Baseline	ΔLER [95% CI]	p	% better	Δr	p	Sites
Baseline	AV static	+0.205 [+0.176, +0.236]	$< 10^{-4}$	100	-0.011	$< 10^{-4}$	92
Baseline	Seasonal sharing	+0.088 [+0.068, +0.109]	$< 10^{-4}$	97	+0.001	0.922	92
Baseline	Phenology rule	+0.038 [+0.025, +0.053]	$< 10^{-4}$	84	+0.001	0.723	92
Baseline	Stress rule	+0.096 [+0.071, +0.124]	$< 10^{-4}$	93	-0.002	0.055	92
+1.5 °C	AV static	+0.208 [+0.180, +0.237]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+1.5 °C	Seasonal sharing	+0.095 [+0.077, +0.114]	$< 10^{-4}$	98	+0.001	0.812	92
+1.5 °C	Phenology rule	+0.033 [+0.022, +0.045]	$< 10^{-4}$	86	+0.002	0.855	92
+1.5 °C	Stress rule	+0.108 [+0.083, +0.134]	$< 10^{-4}$	98	-0.001	0.041	92
+2 °C	AV static	+0.209 [+0.181, +0.238]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+2 °C	Seasonal sharing	+0.097 [+0.079, +0.115]	$< 10^{-4}$	97	+0.001	0.876	92
+2 °C	Phenology rule	+0.032 [+0.022, +0.044]	$< 10^{-4}$	85	+0.002	0.496	92
+2 °C	Stress rule	+0.115 [+0.089, +0.142]	$< 10^{-4}$	98	-0.003	0.016	92
+3 °C	AV static	+0.219 [+0.190, +0.250]	$< 10^{-4}$	100	-0.009	$< 10^{-4}$	92
+3 °C	Seasonal sharing	+0.108 [+0.089, +0.129]	$< 10^{-4}$	96	+0.001	0.935	92
+3 °C	Phenology rule	+0.034 [+0.022, +0.046]	$< 10^{-4}$	86	+0.002	0.330	92
+3 °C	Stress rule	+0.131 [+0.102, +0.162]	$< 10^{-4}$	98	-0.002	0.142	92